
ADVERSARIAL ATTACKS LEVERAGE INTERFERENCE BETWEEN FEATURES IN SUPERPOSITION

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ABSTRACT

The conditions under which adversarial examples (AExs) arise in neural networks (NNs) remain contested: prevailing accounts attribute them either to irregularities in the decision landscape or to sensitivity to non-robust input features. We advance a third, unifying account in which adversarial vulnerability follows from the *efficient* information encoding that NNs adopt in order to represent more features than they have dimensions. Concretely, we show that superposition—the packing of an overcomplete set of features into a lower-dimensional activation space—arranges latent representations so that an adversary can systematically exploit the interference between them. Because the perturbation an attack requires is fixed by this feature geometry, successful attack patterns become predictable from the arrangement of superposed features rather than arbitrary. This mechanistic account reconciles two phenomena that prior explanations leave disconnected: the transferability of attacks between models trained under similar regimes, and the emergence of class-specific vulnerability patterns. In synthetic settings with precisely controlled superposition, we establish—empirically, and through an exact analysis of the linear case—that superposition *suffices* to induce adversarial vulnerability, and we trace the causal pathway that runs from input correlations through feature geometry to the realised perturbations. The same mechanism then governs a vision transformer (ViT) trained on CIFAR-10 once an engineered superposition bottleneck is introduced, showing that the account carries from idealised models to a standard vision architecture and task under a controlled representational intervention. Within these settings, the results recast adversarial vulnerability as an intrinsic corollary of representational compression rather than a defect of the learning process or a fixed property of non-robust inputs, and they motivate defences directed at the semantics of learned representations rather than at input-space norm bounds alone. We release our implementation through an accompanying software repository.

1 INTRODUCTION

Despite more than a decade of intensive study of AExs (Szegedy et al., 2014; Goodfellow et al., 2014), the field has reached no consensus on their underlying cause. This absence of a mechanistic account—an explanation of how minimal, often imperceptible perturbations so drastically alter predictions—carries concrete costs: without it we can neither anticipate which perturbations will succeed, nor explain why attacks transfer across independently trained models, nor design defences on principled rather than purely empirical grounds.

Existing explanations fall into two broad camps (Nakkiran, 2019). The *bug* perspective attributes vulnerability to flaws in learning (Schmidt et al., 2018; Nakkiran, 2019) or to geometric properties of high-dimensional decision boundaries (Fawzi et al., 2016); work in this tradition supplies statistical bounds on quantities such as capacity (Bubeck & Sellke, 2021) or Lipschitz constants (Hein & Andriushchenko, 2017), yet stops short of identifying *which* specific perturbations succeed. The *feature* perspective instead holds that AExs exploit predictive but non-robust statistical regularities in the data (Ilyas et al., 2019), recasting the question as one about the relationship between the data and human perception. Crucially, this view treats non-robust features as fixed properties of the data rather than as artefacts of the encoding strategy that a network adopts. Neither camp articulates how representational constraints and data semantics *jointly* give rise to vulnerability, and it is precisely this interaction that our account makes explicit.

We close this gap with a mechanistic interpretability account in which adversarial vulnerability emerges from the interaction between architectural constraints and data semantics (*i.e.* the human-interpretable properties of the data). We demonstrate that, in this setting, AExs arise from interference between learned representations: adversarial attacks systematically exploit superposed latent structure—the very mechanism through which a network gains additional representational capacity—to assemble perturbations that steer its outputs. This exposes a concrete causal pathway, in which input correlations constrain feature arrangements, those arrangements determine the resulting interference patterns, and the interference patterns in turn dictate both the form and the transferability of attacks. The account is therefore predictive rather than merely descriptive: it specifies which perturbations succeed, and why they transfer between models trained under similar regimes.

Our account builds on two recent theories of how networks encode and process information: the linear representation hypothesis (LRH) (Park et al., 2024) and the theory of superposition (Elhage et al., 2022). The LRH holds that *input features*—the fundamental abstractions of the data—are encoded as linear directions in a network’s representation space. Superposition then permits a network to represent substantially more such features than it has neurons, packing multiple features into shared dimensions and thereby purchasing representational efficiency at the cost of interference, so that perturbing one feature perturbs others in non-obvious ways. We show that precisely this interference constitutes the vulnerability that AExs exploit, and we develop its consequences for understanding a range of adversarial phenomena. Our primary contributions are the following.

- **A unifying mechanistic framework** that reconciles the architectural and feature-based accounts of adversarial vulnerability. In controlled models we establish that data properties induce specific superposition geometries which *directly determine* the realised adversarial perturbations, and that the resulting interference patterns account for both attack transferability and class-specific susceptibility. An exact analysis of the linear case renders this causal pathway precise.
- **Evidence that the mechanism carries to a standard vision architecture**: the same relationships that hold in the toy models reappear, without qualitative change, in a ViT trained on CIFAR-10 once an engineered bottleneck layer places its class representations in superposition.
- **A sharp delineation of the mechanism’s boundary**: superposition is sufficient, but *not necessary*, for adversarial vulnerability. In establishing this boundary we identify *algorithmic brittleness*—a distinct vulnerability mechanism that operates even under strictly orthogonal representations.
- **An actionable consequence**: because our account is mechanistic, a characterisation of the learned representations suffices to construct adversarial attacks directly and gradient-free, which in turn grounds our argument for semantically informed defences.

2 BACKGROUND

We now introduce the two theoretical instruments on which our analysis rests: the LRH and superposition. Formally, let $\mathbf{x} \in \mathcal{X}$ denote the input to a NN and $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ the activation vector of its l -th layer.

Linear representation hypothesis (LRH). The LRH holds that NNs encode many of the variables of their computation—in particular the semantic properties of their inputs—as linear directions in activation space, which then serve as manipulable abstractions for downstream reasoning (Park et al., 2024; Guerner et al., 2023). Two considerations motivate this linear encoding: linear separability allows a network to recognise and manipulate a feature through a single projection, and the dot product with subsequent-layer weights processes such directional features efficiently. A growing body of empirical evidence supports the hypothesis (Gurnee et al., 2023; Park et al., 2024). Let $\mathcal{C} = \{c_1, \dots, c_M\}$ denote a set of M semantically meaningful latent features (*e.g.* concepts such as “presence of shape” or “indoor vs. outdoor”). Formally:

Definition 1 (Linear Representation Hypothesis (LRH)) A neural network layer with activations $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ satisfies the LRH if it represents the latent features $\mathcal{C} = \{c_1, \dots, c_M\}$ as linear directions $\{\mathbf{v}_j\}_{j=1}^M \subset \mathbb{R}^{d_l}$ such that:

$$\mathbf{h}^{(l)}(\mathbf{x}) \approx \sum_{j=1}^M a_j(\mathbf{x}) \mathbf{v}_j$$

where $a_j(\mathbf{x}) \in \mathbb{R}$ represents the activation magnitude of feature c_j , and $\mathbf{v}_j$ is the corresponding linear direction. The features are **linearly accessible** (Costa et al., 2025): inputs $\mathbf{x}_0, \mathbf{x}_1 \in \mathcal{X}$ that

differ mainly in the value of feature c_j while holding other features approximately constant (i.e., $a_j(\mathbf{x}_1) - a_j(\mathbf{x}_0) = k$ and $|a_i(\mathbf{x}_1) - a_i(\mathbf{x}_0)| < \lambda$ for all $i \neq j$ and some small $\lambda > 0$) satisfy:

$$\mathbf{h}^{(l)}(\mathbf{x}_1) - \mathbf{h}^{(l)}(\mathbf{x}_0) \approx k\mathbf{v}_j$$

where k reflects the change in c_j .

Superposition and sparsity. NNs routinely represent far more latent features than they have activation dimensions, $M > d_l$; large language models, for instance, reference many more place names than their residual stream has dimensions. Superposition is the framework that accounts for this regime: a network deploys an overcomplete, non-orthogonal set of feature directions $\{\mathbf{v}_j\}_{j=1}^M$, exploiting two facts. First, a d -dimensional space admits $2^{\Theta(d^2)}$ almost-orthogonal vectors (those with pairwise cosine similarity below ϵ) (Tkocz, 2012); second, sufficiently sparse vectors can be recovered accurately after projection into a lower-dimensional space (Elhage et al., 2022; Bereska & Gavves, 2024; Sawmya et al., 2025). This additional capacity is purchased at the price of two coupled phenomena. First, *polysemanticity* (Scherlis et al., 2022; Lecomte et al., 2023) arises: an individual neuron comes to participate in several distinct features, so that its activation no longer corresponds to a single interpretable concept. Second, and central to our analysis, non-orthogonal feature directions induce *interference*—activating one feature partially activates the others. Networks can suppress these effects only partially, through non-linear operations (e.g. ReLU, softmax) that disambiguate superposed features (Gurnee et al., 2023); the residual interference that survives this suppression is precisely what we show adversaries exploit.

Definition 2 (Superposition Hypothesis) A network layer represents features in superposition if:

1. The number of latent features exceeds layer dimensionality: $M > d_l$
2. There exists non-orthogonal feature directions: $\exists i, j$ with $i \neq j$ such that $\mathbf{v}_i \cdot \mathbf{v}_j \neq 0$
3. Latent feature activations are sparse: $\mathbb{E}_{\mathbf{x}}[\|\mathbf{a}(\mathbf{x})\|_0] \ll M$, where $\mathbf{a}(\mathbf{x}) = [a_1(\mathbf{x}), \dots, a_M(\mathbf{x})]^T$

By packing more features than dimensions, the network purchases representational capacity at the price of feature interference—the very trade-off whose adversarial consequences this paper characterises.

Adversarial attacks. We adopt the standard formalisation of adversarial perturbations and attacks, following (Szegedy et al., 2014; Costa et al., 2024).

Definition 3 (Adversarial attack) Let $\mathbb{B}_p(\mathbf{x}, \epsilon)$ denote the ϵ -ball around $\mathbf{x} \in \mathcal{X}$ in the ℓ_p -sense, e.g. $\mathbb{B}_\infty$ is the ℓ_∞ -ball. Any small δ that p -perturbs $\mathbf{x}$ to create an **adversarial input** $\mathbf{x}' \in \mathbb{B}_p(\mathbf{x}, \epsilon)$ such that $f(\mathbf{x}') \neq f(\mathbf{x})$ (untargeted) or $f(\mathbf{x}') = y_{\text{target}}$ (targeted) is known to be an **adversarial attack**. ℓ_∞ - or ℓ_2 -norm attacks are created via the projected gradient descent (PGD) (detailed in App. A).

Definition 4 (Adversarial robustness) A classifier $g_w(\cdot)$ with parameters w is said to be **robust** at $\mathbf{x}$ with radius $\epsilon > 0$, if $g_w(\mathbf{x}') = g_w(\mathbf{x}) \forall \mathbf{x}' \in \mathbb{B}(\mathbf{x}, \epsilon)$.

3 SUPERPOSITION GEOMETRY DETERMINES ADVERSARIAL ATTACKS

We first establish, in a controlled setting, precisely how adversarial attacks exploit the interference that is rooted in the superposition of latent features. Three questions structure the investigation:

1. Do adversarial perturbations exploit the interference between superposed features?
2. Do correlations in the input shape the geometric arrangement of superposed latent features?
3. Can shared latent geometry explain why attacks transfer between independently trained models?

Experimental setup. To address these questions with the precision they demand, we construct a synthetic classification task that instantiates the established LRH and superposition phenomena (Park et al., 2024; Elhage et al., 2022) under full experimental control. The task is deliberately engineered to satisfy four requirements simultaneously: (1) it furnishes an intuitive classification setting for studying AExs; (2) it represents class concepts as linear directions, as the LRH prescribes; (3) it induces controlled superposition among the resulting latent features; and (4) it preserves *a priori* knowledge of the correspondence between inputs and superposed features, rendering predictions about the adversarial mechanism directly testable.

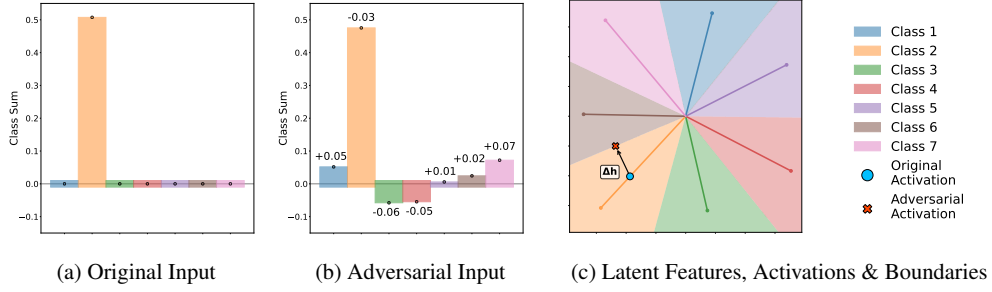


Figure 1: **An adversarial attack exploiting superposition geometry** ($k = 7$, $m = 2$). (a) The original sample. (b) The adversarially perturbed sample: its ground-truth class is unchanged, yet it is misclassified, and the sign and magnitude of each input perturbation are fixed by the configuration of latent representations rather than being arbitrary. (c) The original and adversarial samples in activation space; the arrows denote the column vectors of $\mathbf{W}_e$, *i.e.* the latent representations of the input features. The attack displaces the activation across the decision boundary precisely along the directions dictated by feature interference, exactly as the interference account predicts.

Concretely, we partition the input $\mathbf{x} \in \mathbb{R}^d$ into k equally sized groups $\mathbf{x} = [\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)}]$, where $\mathbf{x}^{(j)}$ collects the p input features associated with class j . The label identifies the group with the largest sum, $y = \operatorname{argmax}_{j \in \{1, \dots, k\}} \sum_{i=1}^p x_i^{(j)}$, so that each entry $x_i^{(j)}$ constitutes evidence for class j and the p features within a group encode its distinct class attributes. Following Elhage et al. (2022), we sample $x_i^{(j)} \sim \text{Uniform}(0, 1)$ under sparsity level S (the probability that an entry is set to zero).

A two-layer network compresses the input into m dimensions through a bottleneck encoder, $\mathbf{h} = \text{ReLU}(\mathbf{W}_e \mathbf{x} + \mathbf{b}_e) \in \mathbb{R}^m$ (with $m < k < d$), and reconstructs the class scores through a decoder, $\mathbf{z} = \mathbf{W}_d \mathbf{h} + \mathbf{b}_d \in \mathbb{R}^k$. Our primary configuration employs cross entropy (CE) loss without ReLUs or biases; we adopt it for its analytic transparency, noting that the softmax already limits interference by exponentially suppressing contributions from interfering latent features. To confirm that our conclusions are not artefacts of this configuration, we replicate them in a second setup that uses mean squared error (MSE) loss with ReLUs and biases. In both cases the representational challenge is identical: compress $d = k \times p$ input dimensions into $m < k$ latent dimensions, and then recover enough information to identify the class with the largest sum.

Whereas naturalistic tasks encode nuanced concepts that must first be discovered and interpreted, this construction supplies ground-truth class labels for the latent features by design. We can therefore recompute the true class after any perturbation directly from the group sums, which lets us define an AEx through two conditions that must hold jointly:

1. the model’s prediction changes under perturbation (*e.g.* from class A to B); and
2. the true class is preserved (*e.g.* still class A).

This criterion cleanly separates genuine AExs from perturbations that in fact alter the ground truth (*i.e.* change which group attains the highest sum). Attacks are generated using ℓ_∞ - and ℓ_2 -norm PGD.

Evaluation metrics. We interpret the columns of the encoder weight matrix $\mathbf{W}_e$ as the latent representations $\mathbf{v}_i$ of the input features x_i . Because input feature $x_i^{(j)}$ contributes exclusively to class j , the directions $\mathbf{v}_i$ for $i \in \text{class } j$ are mutually aligned. This exact correspondence between input features and class-aligned latent directions is what allows us to trace, without ambiguity, how a perturbation to the inputs $\mathbf{x}^{(j)}$ of a specific class translates into movement in latent space. For each successful PGD-generated AEx $\mathbf{x}_{adv} = \mathbf{x}_{orig} + \delta$, we measure the following.

1. **Input perturbation profile (IPP):** The sign and magnitude of each δ_i , describing how different input features are perturbed in an attack.
2. **Latent attack alignment:** The similarity $\Delta \mathbf{h} \cdot \mathbf{v}_c$ between the *latent attack vector* $\Delta \mathbf{h} = \mathbf{h}_{adv} - \mathbf{h}_{orig}$ and class c ’s representation.
3. **Attack transferability:** Success rate of attacks generated on one model when applied to another.
4. **Robust accuracy:** The fraction of examples that remain correctly classified after perturbation δ with $\|\delta\|_p \leq \epsilon$ to a sample of inputs.

3.1 RESULTS

Do attacks exploit interference between latent features in superposition? Changing a sample’s class requires moving it across a decision boundary, but what determines the input perturbation that achieves this displacement? Two prevailing intuitions make opposing predictions. The *feature intuition* anticipates that an attack simply amplifies the input features of the target class; the *bug intuition* holds instead that adversarial perturbations modify inputs in unpredictable ways, so that knowledge of the feature representations offers no purchase on them. Our results support neither, and instead vindicate a third account.

Fig. 1 presents a representative AEx in our setting: an input of Class 2 (orange) and its perturbed counterpart (annotated with its IPP), which is misclassified as Class 6 (brown). The perturbation does not simply increase the target-class features, directly contradicting the feature intuition; yet it is equally far from random. It obeys a precise, systematic correspondence between δ and the configuration of latent features—the signature of the third account that the remainder of this section establishes: *attacks are mediated by latent geometry*.

Quantitatively, the IPP is tightly coupled to the latent attack alignment. The sign and magnitude of the perturbation $\delta^{(j)}$ track how that class’s latent representation $\mathbf{v}^{(j)}$ aligns with the latent attack vector $\Delta\mathbf{h}$: positively aligned representations ($\mathbf{v}^{(j)} \cdot \Delta\mathbf{h} > 0$) have their input features proportionally amplified, whereas negatively aligned representations are attenuated. This is precisely the behaviour visible in Fig. 1(b)—input perturbations set by the dot products in Fig. 1(c) so as to displace the sample across the decision boundary.

To determine whether PGD attacks *specifically* exploit superposition geometry, we compare them against the theoretically optimal interference-exploiting perturbations derived in Sec. 3.2. For each configuration (k, m) we train

5 models from distinct random seeds, generate 1000 PGD attacks per model, and compute the cosine similarity between each successful perturbation and the corresponding optimum, discarding perturbations with ℓ_2 norm below ϵ so as to retain only meaningful attacks; a random baseline is constructed from norm-matched perturbations compared against the same optimum. As Tab. 1 reports, PGD aligns near-perfectly with the theoretical optimum across every dimensionality tested (cosine similarity 0.92–0.97, against ≈ 0 for the random baseline), and one-sample t -tests against that baseline yield $p < 10^{-10}$ in all three configurations. Because the optimum is by construction the interference-exploiting perturbation, and PGD recovers it almost exactly, we conclude that PGD attacks operate by exploiting superposition interference.

Finding: Adversarial attacks systematically exploit interference between superposed features. Successful PGD attacks are predictable given the specific superposition geometry, rather than random.

Do input correlations determine latent feature geometry? Having shown that attacks are governed by feature geometry, we now ask what determines that geometry. We hypothesise that correlations in the input data fix the geometric arrangement of the latent features, and we test this by systematically manipulating the correlation structure of the training data and observing how it constrains the degrees of freedom in the learned representations. Three correlation patterns span a range of constraint strengths. Under the *uncorrelated* condition, features are sampled independently, exactly as in the previous subsection. Under *paired* correlations, each feature in class i is coupled to a feature in class j , so that activating one activates its partner. Under *global* correlations, each sample carries a random phase that induces a cyclic activation pattern: classes adjacent in index share similar activation probabilities and therefore co-activate frequently, whereas widely separated classes co-activate only rarely.

Fig. 2 shows how the correlation structure governs the arrangement of the $\mathbf{v}^{(j)}$. We quantify geometric similarity by forming, for each model, the matrix of pairwise cosine similarities over all feature pairs, and then correlating these matrices across random seeds, so that a higher correlation denotes a more consistent geometric arrangement. Evaluating 25 seed pairs per condition in the $m = 2, k = 7$ setup yields geometric similarities of 0.23 ± 0.14 (uncorrelated), 0.47 ± 0.32 (paired), and 0.88 ± 0.06 (global) (mean $\pm$ std)—a clear monotonic ordering. Uncorrelated data admits highly variable geometries across seeds, paired correlations partially pin the arrangement, and global correlations force near-identical geometries. Pairwise two-sample t -tests between all three conditions, Bonferroni-corrected ($\alpha = 0.05$ over 3 comparisons), confirm the separation at $p < 10^{-4}$ throughout, and the same ordering recurs across the range of (k, m) configurations reported in App. C.3.

The governing mechanism is *interference avoidance*: features that co-activate frequently are arranged so as to minimise their mutual interference, and the strength of the correlations sets the number of constraints imposed on this arrangement. Uncorrelated data permit any arrangement that separates the classes; paired correlations fix the relative positions of coupled features; and global correlations pin down a unique geometry up to rotation.

Finding: Input correlations constrain feature geometry. Stronger correlations reduce geometric degrees of freedom, forcing different initialisations to converge to similar arrangements.

Does shared geometry explain attack transferability? Having shown that input correlations induce consistent latent geometries across initialisations, we now test the prediction that follows: if an attack is tailored to a specific interference pattern, it should transfer successfully only between models that share a geometric arrangement, and transferability should therefore be a direct function of geometric similarity.

Reusing the seed pairs from the previous paragraph, we generate attacks on source models and evaluate their success on target models within each correlation condition (25 transfer measurements per condition), with every model exceeding 95% clean accuracy. Transfer rates track geometric similarity precisely. In the $k = 7, m = 2$ configuration (Fig. 2), globally correlated data yield $94\% \pm 12\%$ transfer, paired correlations an intermediate $32\% \pm 20\%$, and uncorrelated data only $18\% \pm 9\%$ (mean $\pm$ std)—a fivefold spread that mirrors the geometric-similarity ordering established above.

The underlying account is mechanistic. Each perturbation component amplifies or suppresses a feature, and these effects combine constructively to drive the representation across a decision boundary. Applied to a model with a different feature arrangement, the same components no longer align: contributions that previously interfered constructively now cancel or interfere destructively, and the attack fails. Shared interference geometry thus emerges as the controlling factor for transferability in this setting.

Finding: Attack transferability is governed by shared interference patterns.

Does reducing superposition suppress these attacks? If superposition creates vulnerability through interference, then removing the superposition should remove the vulnerability. Three complementary interventions confirm this prediction, and jointly isolate interference—rather than reduced dimensionality or classification

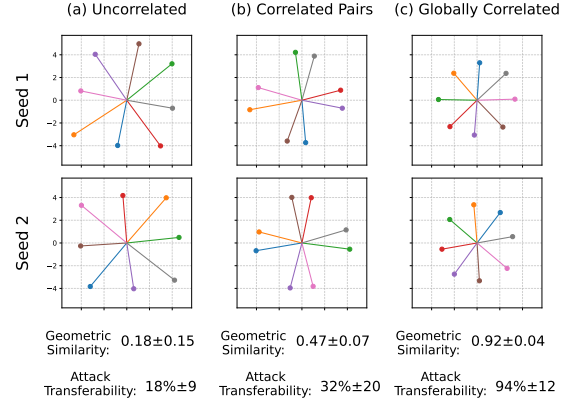


Figure 2: Stronger input correlations yield more consistent feature geometries across initialisations, driving attack transferability from 18% (uncorrelated) to 94% (global) in the $k = 7, m = 2$ configuration—a monotonic, fivefold increase. Error bars denote standard deviations.

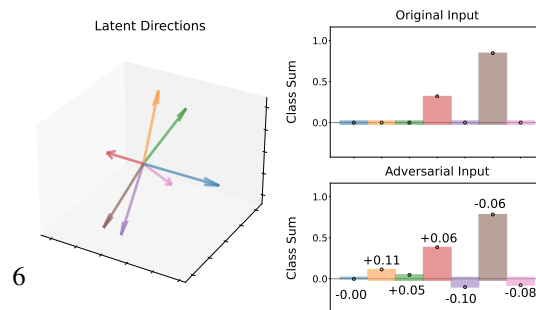


Figure 3: An adversarial attack (from class 5 to

difficulty—as its cause. *First*, setting $m = k$ forces the network to represent each class direction $\mathbf{v}^{(j)}$ orthogonally; in this regime any perturbation that changes the prediction must necessarily change the ground truth, since moving a sample from class A to class B requires making the class-B feature sum exceed the class-A sum—genuinely transforming the sample. Consistent with this, we observe *zero* successful adversarial examples across 1000 attempts at every ϵ tested. *Second*, to disentangle superposition from raw capacity, we hold m fixed and vary k : robust accuracy falls monotonically with the superposition pressure k/m (App. Tab. 5), demonstrating that vulnerability scales with the degree of interference rather than with capacity *per se*. *Third*, to verify that attacks concentrate on superposed features, we constrain a single class’s latent vector $\mathbf{v}^{(\perp)}$ to remain orthogonal to all others throughout training; when attacking between classes that themselves remain in superposition (e.g. class a to b with $\mathbf{v}_a \not\perp \mathbf{v}_b$), the inputs of the orthogonal class are left entirely unperturbed (Fig. 3). The three interventions converge on a single conclusion: interference between superposed features, and not reduced representational capacity, is the operative cause of the vulnerability.

Finding: Adversarial attacks use their budget to exploit those features in superposition.

3.2 FORMAL ANALYSIS

We now place the empirical observations on an exact footing. In the linear setting, the relationship between superposition interference and adversarial vulnerability admits a closed-form derivation, which elevates the preceding correlations to causal statements. We retain the notation of the experimental setup: $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ induces superposition through its columns $\mathbf{v}_i$, the decoder is the encoder transpose (as observed empirically), and the logit for class j is $z_j = \mathbf{v}_j^\top \mathbf{h}$.

Proposition 1. *The optimal input perturbations δ that maximise movement from class j to class k under constraint $\|\delta\|_2 = \epsilon$ satisfy $\delta \propto \mathbf{W}_e^\top \mathbf{n}$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is the normal to the decision boundary between classes.*

Proof Sketch. To move from class j to k , we maximise the logit margin $z_k - z_j = \mathbf{n}^\top (\mathbf{h} + \Delta \mathbf{h})$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ and $\Delta \mathbf{h} = \mathbf{W}_e \delta$. This becomes $\max_{\|\delta\|_2 = \epsilon} \delta^\top \mathbf{W}_e^\top \mathbf{n}$. By Cauchy-Schwarz, this is maximised when $\delta \propto \mathbf{W}_e^\top \mathbf{n}$. Full proof in App. B. $\square$

Corollary 1 (Interference Drives Vulnerability) *The adversarial perturbation magnitude for feature i is $|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$, directly proportional to the differential interference between feature i and the class representations.*

This corollary makes explicit how superposition creates adversarial vulnerability: each input feature i is perturbed in proportion to its differential interference with the class representations. Under superposition, non-orthogonality guarantees that even semantically unrelated features interfere with the class decision, and the adversarial perturbation exploits precisely these dependencies to steer the output.

Proposition 2. *Models with feature representations related by orthogonal transformation $\mathbf{Q}$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$) share identical optimal attack directions in input space.*

Proof Sketch. Under transformation $\mathbf{v}'_i = \mathbf{Q} \mathbf{v}_i$, optimal perturbations remain invariant: $\delta'_i \propto (\mathbf{v}'_i)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = \mathbf{v}_i^\top \mathbf{Q}^\top \mathbf{Q} (\mathbf{v}_k - \mathbf{v}_j) = \mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j) = \delta_i$. Full proof in App. B. $\square$

In combination with the empirical results, these propositions establish a complete **mechanistic pathway** from data statistics to attack behaviour: (i) input correlations constrain the feature arrangements realised in superposition; (ii) these geometric arrangements determine the interference patterns; (iii) the interference patterns dictate the optimal perturbations through $\delta \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$ (Proposition 1); and (iv) shared geometric constraints induce shared interference patterns across models, which is precisely what enables transferability (Proposition 2):

Correlations $\xrightarrow{\text{constrain}}$ **Feature Geometry** $\xrightarrow{\text{determines}}$ **Interference Patterns** $\xrightarrow{\text{enable}}$ **Transferability**

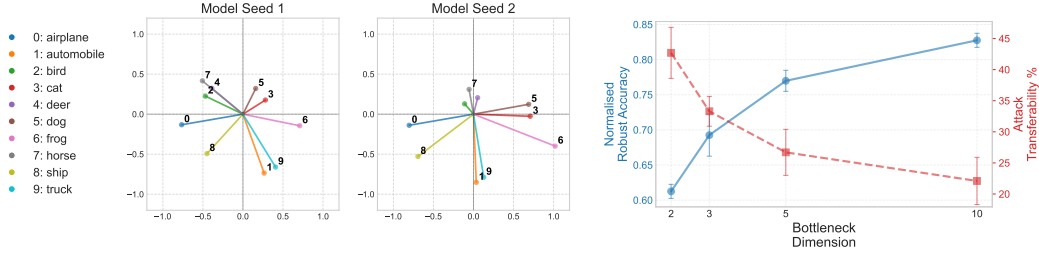


Figure 4: Left: the geometric structure of the CIFAR-10 class representations is preserved across models trained from different seeds. Right: attack transferability and normalised robust accuracy as a function of the bottleneck dimension, showing that transferability rises and (capacity-normalised) robustness falls as superposition pressure grows—the same joint dependence established for the toy models.

4 ATTACKS IN A STANDARD VISION ARCHITECTURE

We now show that the mechanism identified in the controlled setting governs a standard vision architecture and task: a ViT (Dosovitskiy et al., 2020) trained on CIFAR-10 (Krizhevsky, 2009), with the same conclusions replicated for a ResNet-18 (He et al., 2016) in App. D.2.1. The transfer is not automatic: we induce it by inserting a bottleneck layer immediately before the classification output, which forces the latent class features into superposition and thereby recreates, within a realistic backbone, the representational regime analysed in Sec. 3. This engineered bottleneck is a deliberate intervention—the point is to test whether the toy-model relationships persist once class features are placed in superposition inside a full vision model, rather than to claim that such superposition arises unbidden in an off-the-shelf classifier. Within this setting we examine three quantities in turn: (i) how the degree of superposition affects attack transferability and robustness; (ii) whether a consistent arrangement of the superposed representations emerges across initialisations; and (iii) whether the attacks exploit feature interference.

Setup. We train the ViT (6 layers, patch size 4, residual stream $d = 512$) on CIFAR-10 across five random seeds. After training, we replace the original classification head with a bottleneck: a linear encoder that compresses the pre-classification activations into an m -dimensional space, followed by a linear decoder that expands them back to the $k = 10$ CIFAR-10 classes. Trained with CE loss and no ReLU—deliberately mirroring the configuration of Sec. 3—this bottleneck forces the k class representations into m -dimensional superposition. The base ViT weights are frozen while the bottleneck is trained, and we sweep the bottleneck dimension $m \in \{2, 3, 5, 10\}$ to modulate the degree of superpositional compression. AExs are generated for each bottlenecked model with both ℓ_∞ - and ℓ_2 -norm PGD, using a small step size and many iterations, and terminating once the prediction flips (full details in App. D.1). Crucially, robust accuracy is normalised by each model’s clean accuracy, so that the comparison across bottleneck dimensions controls for the differing capacities they afford; attack transferability is measured exactly as in Sec. 3.

Results. The base models attain a mean clean accuracy of 81%, and, as expected, the bottleneck degrades clean accuracy monotonically with increasing compression: 54%, 67%, 76%, and 81% for $m = \{2, 3, 5, 10\}$, respectively.

Three observations reproduce the toy-model findings and thereby demonstrate that the mechanism carries to this realistic setting. First, Fig. 4 (right) shows that as the bottleneck m shrinks—*i.e.* as superposition pressure rises—normalised robust accuracy falls while attack transferability across initialisations rises. Because robust accuracy is normalised by clean accuracy, this decline cannot be attributed to the reduced capacity of the smaller bottleneck alone; it follows instead from the denser packing of the class representations, which leaves less room to separate them, so that small input perturbations induce widespread interference across the compressed features. The concurrent rise in transferability is precisely what the analysis of Sec. 3 predicts: greater superposition reduces the degrees of freedom available to the feature geometry, so that a more constrained representational space affords the network fewer viable arrangements of its class features, drives independent initialisations toward more similar geometries, and hence toward the shared interference patterns that make attacks transfer. App. D.1.1 confirms this trend across the full range of perturbation magnitudes.

The second observation corroborates this picture: Fig. 4 (left) shows that the m -dimensional class representations within the bottleneck exhibit a consistent ordering and clustering across models. As in Sec. 3, this consistency indicates that inherent correlations—here between class representations rather than raw inputs—guide the formation of the superposed geometry; moreover, the resulting groupings are semantically meaningful, with related classes (e.g. ‘cat’ and ‘dog’) clustering together. Third, when AExs generated for a bottlenecked model are passed through the corresponding base ViT, they produce concordant relative logit changes ($\text{logit}_{\text{base}}(\mathbf{x}_{\text{adv}}) - \text{logit}_{\text{base}}(\mathbf{x}_{\text{orig}})$), establishing that the two models respond consistently to the same perturbations and that those perturbations are not mere artefacts of the bottleneck; the added compression is nonetheless what converts this shared response into an outright misclassification.

5 ADVERSARIAL VULNERABILITY VIA ALGORITHMIC BRITTLINESS

The preceding sections establish that feature superposition *suffices* for adversarial vulnerability. We now delimit the mechanism from the opposite side, asking whether it is also *necessary*: we study a setting in which features are represented orthogonally rather than in superposition, and show that vulnerability nonetheless persists. This delineates the boundary of the superposition account and, in doing so, substantiates the paper’s broader thesis—that the analysis of feature representations is a general instrument for reasoning about robustness. We pursue this through three questions: *how do such attacks operate mechanistically, can this understanding be turned directly into gradient-free attacks, and does naive robustness certification alter the underlying failure mode or merely raise the attack budget it requires?*

Setup. We train a linear encoder followed by an MLP on one-hot integers (a, b) to predict $(a + b) \bmod P$. As detailed in App. E and Nanda et al. (2023), the network discovers a trigonometric algorithm for this task. As part of this learned algorithm¹ the encoder projects inputs onto non-basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at key frequencies $\omega_k = 2\pi k/P$, and the MLP then composes $\cos((a + b)\omega_k)$ through sum-angle identities to attain 100% accuracy. Crucially, these features are represented *orthogonally*—explicitly not in superposition. We generate AExs by relaxing the one-hot constraint and perturbing only input a with PGD, thereby displacing inputs off the training manifold.

Attacks exploit the learned key frequencies. Fourier analysis of the successful perturbations δ (App. Fig. 6) reveals that their dominant frequencies coincide *exactly* with the learned algorithm’s key frequencies ω_k . This frequency-specificity is decisive: 0% of attacks transfer to models with different ω_k (e.g. those trained from different seeds), which demonstrates that the attack is keyed to the model’s particular learned frequencies rather than to any generic input direction. Mechanistically, the frequency-aligned perturbation induces misclassification by corrupting the sinusoidal values that the MLP reads off the encoder features: δ converts the one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$, and the contributions from the now non-zero entries perturb those sinusoidal values, disrupting the MLP’s downstream computation. Because the algorithm relies on the constructive interference of every ω_k at the logits, perturbing a single ω_k is insufficient; a successful δ must maximally alter several frequency components at once. App. E.1 develops the detailed intuition for this mechanism.

Analytic construction of attacks. The identified mechanism is directly actionable: it allows us to construct gradient-free, *informed* perturbations as composite sinusoidal waves at the matching ω_k . On $0 + 0 \bmod P$, these hand-built attacks succeed at $\|\delta\|_2 = 0.22$ —within $\approx 1.3\times$ of the PGD budget (0.17), and far below every naive alternative (single-index 0.97, ℓ_∞ noise 2.04, uniform 6.20). That a perturbation constructed *purely* from the learned frequencies nearly matches gradient-based PGD is strong evidence that we have identified the correct mechanism; it further shows that a representation-aware view of vulnerability yields concrete attack recipes rather than merely post-hoc rationalisations.

Do defences mitigate this algorithmic vulnerability? We apply certified training against ℓ_∞ perturbations (Müller et al., 2023). At fixed attack budgets, *robust* accuracy improves markedly from the pre- to the post-robustified model: 0% $\rightarrow$ 85% at $\epsilon_\infty = 5e-3$, 0% $\rightarrow$ 96% at $2e-3$, and 76% $\rightarrow$ 100% at $1e-3$. These gains are, however, superficial: attacked afresh with PGD, the certified

¹We refer to the specific computation the network executes to solve the task as the *model’s learned algorithm*. Because this algorithm depends on extracting precise values from learned feature representations (frequencies), it remains *brittle* under small perturbations to the bases; we term this the *algorithmic brittleness*.

model fails through the *identical* frequency-based brittleness, merely at larger perturbation norms. Certification thus displaces the required budget without addressing the underlying fragility—a finding that directly motivates defences targeting the identified semantics (here, frequency manipulation) rather than relying on broad input-space norm bounds alone.

Relation to the paper’s thesis. This section contributes two points to the paper’s broader argument. First, it establishes that superposition is *sufficient but not necessary* for adversarial vulnerability: orthogonal representations remain attackable through a distinct mechanism, algorithmic brittleness, which fixes the precise boundary of the superposition account developed above. Second, it demonstrates that viewing AExs through the lens of *architecture–semantics interaction*—the common thread across all our settings—is actionable in both directions, guiding the construction of attacks (via representation-informed perturbations) and the design of defences alike.

Finally, and though secondary to the present analysis, the two mechanisms share a dependence on sparsity assumptions: an upstream violation of sparsity (*e.g.* noise, or superposed features) can hand a downstream orthogonal circuit an effectively ‘pre-perturbed’ input, activating the same frequency-based failure. This link—which begins to reunite the two mechanisms under a single representational account—we develop as the reconciliation conjecture in App. E.

6 RELATED WORK

Superposition & latent representations. Research on superposition has characterised several feature-interference mechanisms: Nanda (2023) distinguish *representational* from *computational* superposition, and Gurnee et al. (2023) separate *alternating* from *simultaneous* interference. A recurring theme is that data correlations shape feature arrangements: correlated feature pairs become orthogonal (Elhage et al., 2022), correlations drive the formation of superposition (Chan, 2024), and they induce semantic clustering (Prieto et al., 2025a). Our results build directly on these observations, extending them by rendering the learned arrangement *predictive* of adversarial behaviour. Beyond superposition, representation geometry is further shaped by spectral bias (Rahaman et al., 2019), neural collapse (Kothapalli, 2023), and the optimisation objective (Casper, 2023), while sparse autoencoders (SAEs) have been employed to disentangle features held in superposition within vision models (Lim et al., 2024; Joseph et al., 2025).

Adversarial vulnerability. This literature is organised around an attack–defence dichotomy. Attacks (Szegedy et al., 2014) apply imperceptible perturbations so as to maximise model error—notably FGSM (Goodfellow et al., 2014) and PGD (Madry et al., 2018)—while adversarial robustness denotes the maintenance of performance under such attacks. Defences range from adversarial training (Madry et al., 2018) to certified training (Wong & Kolter, 2018; Palma et al., 2024), the latter furnishing guarantees against every perturbation within a given input bound.

The role of input & latent features on adversarial robustness. Vulnerability has been attributed to predictive but brittle ‘non-robust’ input features (Ilyas et al., 2019; Goh, 2019) and to learning shortcuts (Li et al., 2023), with perturbation characteristics known to vary by dataset (Maiya et al., 2021). A possible link between superposition and AExs was suggested by Elhage et al. (2022) and subsequently contested by Casper (2023); within the controlled and engineered settings we study, our analysis resolves this dispute in the affirmative, exhibiting the precise interference mechanism and placing it on a formal footing. Relatedly, a tension between representational efficiency and adversarial robustness has been observed (Barsbey et al., 2025); Zhang et al. (2021) show that adversarial perturbations push latent representations across decision boundaries; and Ganeshan et al. (2019) find that PGD targets final layers and accordingly propose feature-based attacks. Finally, transferability is known to be limited by representational discrepancies between models (Li et al., 2023; Wang et al., 2024), and decorrelating features reduces transfer (Wiedeman & Wang, 2022)—a result our mechanistic account explains directly, since decorrelation destroys the shared interference geometry on which transfer depends.

7 DISCUSSION & CONCLUDING REMARKS

We have established that adversarial attacks exploit the interference patterns that arise from the geometry of superposed features in NNs. Across controlled toy models and an engineered-bottleneck ViT, our experiments show that data properties—specifically correlations and sparsity—induce distinct superposition geometries, and that these geometries in turn create predictable adversarial

vulnerabilities. Within these settings the geometric arrangement and its attendant interference suffice to predict phenomena that prior accounts leave disconnected, including attack transferability and class-specific susceptibility, and the linear analysis renders the underlying relationship exact rather than merely correlational. Taken together, the results reframe adversarial vulnerability as an intrinsic corollary of the way networks efficiently encode large numbers of features through superposition, rather than as a defect of the learning process or a fixed property of the inputs.

Limitations & future work. We are deliberate about the boundary of what these results establish. Our conclusions are drawn from controlled toy models and from a ViT in which the class features are placed in *engineered* superposition; the analysis is, moreover, primarily linear or near-linear around a bottleneck representation. Naturally trained large-scale networks instead involve interference among many unknown, unlabelled features distributed across multiple layers, together with nonlinear, spatial, and optimisation effects that our theory does not attempt to capture. We therefore do not claim that superposition interference is the dominant or universal cause of adversarial examples in ordinary large models; what we establish is that it *suffices*, that in the regimes we study it *predicts* attack directions, transferability, and class-specific susceptibility, and that a representation-level analysis is a productive instrument for reasoning about robustness. Consistent with the broader adversarial-robustness literature, the precise vulnerability mechanism should be expected to vary with attack and model type: constrained attacks such as universal perturbations, jailbreaks, and patch attacks, for instance, may be unable to activate the specific feature combinations that superposition interference requires. These boundaries define, rather than diminish, the scope of the account.

Several directions follow directly from this analysis: quantifying the ‘adversarial cost’ of superposition, relating it to the accuracy–robustness trade-off, and characterising how robust training reshapes superposition geometry. Extending the account to the unlabelled, multi-layer features of large-scale networks—and thereby testing whether the interference geometry identified here can be *detected*, rather than imposed, in naturally trained models—is the central open problem; to this end we plan to employ SAEs (Bricken et al., 2023). The exploratory results in App. F, though at present inconclusive, indicate that this is a feasible path rather than an established finding.

ETHICS STATEMENT

Given its investigative and mechanistic character, our framework is not anticipated to carry specific negative impacts or immediate societal consequences. The experiments rely solely on publicly available data (CIFAR-10) and synthetic toy models, and involve neither human subjects nor personally identifiable information. The gradient-based attacks we employ are already well established in the literature; the work therefore introduces no novel attack capability, but rather a theoretical understanding of existing phenomena. We declare no conflicts of interest. By clarifying the relationship between superposition and adversarial vulnerability, we aim to inform research that balances model capability against robustness, and thereby to contribute to the development of more trustworthy AI systems.

REPRODUCIBILITY STATEMENT

Source code for all experiments is included in the supplementary materials. The toy-model experiments (Sec. 3) are specified in full—architecture, training procedure, evaluation metrics, and data-generation process. The ViT experiments (Sec. 4) report their architecture and training procedure, elaborated in App. D.1, while the modular-arithmetic experiments (Sec. 5) detail the network architecture, data encoding, and learned trigonometric algorithm, with further specifics in App. E.2. All experiments were conducted on an AMD Ryzen Threadripper 3970X 32-core processor with 256GB RAM and an RTX 3090 GPU. The additional ResNet-18 and SAE experiments are documented in App. D.2.1 and App. F, respectively.

LLM USAGE STATEMENT

This work used LLMs to assist with prior-literature search, to automate the generation of LaTeX tables from CSV files (with all numerical values independently verified), and to suggest alternative phrasings that improve the clarity of the manuscript. All substantive research contributions and conclusions remain the original work of the authors.

REFERENCES

- Melih Barsbey, Antônio H Ribeiro, Umut Şimşekli, and Tolga Birdal. On the interaction of compressibility and adversarial robustness. *arXiv preprint arXiv:2507.17725*, 2025.
- Leonard Bereska and Stratis Gavves. Mechanistic interpretability for AI safety - a review. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.
- Sébastien Bubeck and Mark Sellke. A universal law of robustness via isoperimetry. In Marc’Aurelio Ranzato, Alina Beygelzimer, Yann N. Dauphin, Percy Liang, and Jennifer Wortman Vaughan (eds.), *Advances in Neural Information Processing Systems*, pp. 28811–28822, 2021.
- Stephen Casper. EIS IX: Interpretability and adversaries. *AI Alignment Forum*, February 2023.
- Lawrence Chan. Superposition is not “just” neuron polysemanticity. *AI Alignment Forum*, April 2024.
- Joana C Costa, Tiago Roxo, Hugo Proença, and Pedro Ricardo Morais Inacio. How deep learning sees the world: A survey on adversarial attacks & defenses. *IEEE Access*, 12:61113–61136, 2024.
- Valérie Costa, Thomas Fel, Ekdeep Singh Lubana, Bahareh Tolooshams, and Demba Ba. From flat to hierarchical: Extracting sparse representations with matching pursuit. *arXiv preprint arXiv:2506.03093*, 2025.

648 Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hier-
649 archical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*,
650 pp. 248–255, 2009.

651 Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas
652 Unterthiner, Mostafa Dehghani, Matthias Minderer, G Heigold, S Gelly, et al. An image is
653 worth 16x16 words: Transformers for image recognition at scale. In *International Conference on*
654 *Learning Representations*, 2020.

656 Rüdiger Ehlers. Formal verification of piece-wise linear feed-forward neural networks. In Deepak
657 D’Souza and K. Narayan Kumar (eds.), *Automated Technology for Verification and Analysis -*
658 *15th International Symposium, ATVA 2017, Pune, India, October 3-6, 2017, Proceedings*, volume
659 10482 of *Lecture Notes in Computer Science*, pp. 269–286. Springer, 2017.

660 Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec,
661 Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, et al. Toy models of superposition.
662 *arXiv preprint arXiv:2209.10652*, 2022.

664 Alhussein Fawzi, Seyed-Mohsen Moosavi-Dezfooli, and Pascal Frossard. Robustness of classifiers:
665 from adversarial to random noise. *Advances in neural information processing systems*, 29, 2016.

666 Aditya Ganeshan, Vivek BS, and R Venkatesh Babu. FDA: Feature disruptive attack. In *Proceedings*
667 *of the IEEE/CVF International Conference on Computer Vision*, pp. 8069–8079, 2019.

669 Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever,
670 Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. In *The Thirteenth*
671 *International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*,
672 2025.

673 Gabriel Goh. A discussion of ‘adversarial examples are not bugs, they are features’: Two examples
674 of useful, non-robust features. *Distill*, 2019.

676 Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial
677 examples. *arXiv preprint arXiv:1412.6572*, 2014.

678 Clément Guerner, Anej Svete, Tianyu Liu, Alexander Warstadt, and Ryan Cotterell. A geometric
679 notion of causal probing. *arXiv preprint arXiv:2307.15054*, 2023.

681 Wes Gurnee, Neel Nanda, Matthew Pauly, Katherine Harvey, Dmitrii Troitskii, and Dimitris Bertsimas.
682 Finding neurons in a haystack: Case studies with sparse probing. *Trans. Mach. Learn. Res.*, 2023.

683 Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image
684 recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*,
685 pp. 770–778, 2016.

687 Matthias Hein and Maksym Andriushchenko. Formal guarantees on the robustness of a classi-
688 fier against adversarial manipulation. In *Advances in Neural Information Processing Systems*,
689 volume 30, 2017.

690 Patrick Henriksen and Alessio Lomuscio. Robust training of neural networks against bias field
691 perturbations. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(12):14865–14873,
692 Jun. 2023.

694 Nikolaus Howe, Ian McKenzie, Oskar Hollinsworth, Michał Zajac, Tom Tseng, Aaron Tucker,
695 Pierre-Luc Bacon, and Adam Gleave. Scaling trends in language model robustness, 2025.

696 A. Ilyas, S. Santurkar, D. Tsipras, L. Engstrom, B. Tran, and A. Madry. Adversarial examples are not
697 bugs, they are features. In *Advances in Neural Information Processing Systems 32*, pp. 125–136,
698 2019.

699 Sonia Joseph, Praneet Suresh, Lorenz Hufe, Edward Stevinson, Robert Graham, Yash Vadi, Danilo
700 Bzdok, Sebastian Lapuschkin, Lee Sharkey, and Blake Aaron Richards. Prisma: An open source
701 toolkit for mechanistic interpretability in vision and video, 2025.

- Guy Katz, Clark W. Barrett, David L. Dill, Kyle Julian, and Mykel J. Kochenderfer. Reluplex: An efficient SMT solver for verifying deep neural networks. In Rupak Majumdar and Viktor Kuncak (eds.), *Computer Aided Verification - 29th International Conference*, volume 10426, pp. 97–117. Springer, 2017.
- Vignesh Kothapalli. Neural collapse: A review on modelling principles and generalization. *Trans. Mach. Learn. Res.*, 2023.
- Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. CIFAR-10 dataset.
- Victor Lecomte, Kushal Thaman, Rylan Schaeffer, Naomi Bashkansky, Trevor Chow, and Sanmi Koyejo. What causes polysemanticity? an alternative origin story of mixed selectivity from incidental causes. *arXiv preprint arXiv:2312.03096*, 2023.
- Ang Li, Yifei Wang, Yiwen Guo, and Yisen Wang. Adversarial examples are not real features. *Advances in Neural Information Processing Systems*, 2023.
- Hyesu Lim, Jinho Choi, Jaegul Choo, and Steffen Schneider. Sparse autoencoders reveal selective remapping of visual concepts during adaptation. *arXiv preprint arXiv:2412.05276*, 2024.
- Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. On the biology of a large language model. *Transformer Circuits Thread*, 2025.
- Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards Deep Learning Models Resistant to Adversarial Attacks. In *International Conference on Learning Representations*, February 2018.
- Shishira R. Maiya, Max Ehrlich, Vatsal Agarwal, Ser-Nam Lim, Tom Goldstein, and Abhinav Shrivastava. A Frequency Perspective of Adversarial Robustness. *arXiv*, 2021.
- Mark Niklas Müller, Franziska Eckert, Marc Fischer, and Martin T. Vechev. Certified training: Small boxes are all you need. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.
- Preetum Nakkiran. A discussion of ‘adversarial examples are not bugs, they are features’: Adversarial examples are just bugs, too. *Distill*, 2019.
- Neel Nanda. A comprehensive mechanistic interpretability explainer and glossary, 2023.
- Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.
- Alessandro De Palma, Rudy Bunel, Krishnamurthy Dvijotham, M. Pawan Kumar, and Robert Stanforth. IBP regularization for verified adversarial robustness via branch-and-bound. *CoRR*, abs/2206.14772, 2022.
- Alessandro De Palma, Rudy Bunel, Krishnamurthy (Dj) Dvijotham, M. Pawan Kumar, Robert Stanforth, and Alessio Lomuscio. Expressive losses for verified robustness via convex combinations. In *The Twelfth International Conference on Learning Representations, ICLR, 2024*.
- Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net, 2024.
- L Prieto, E Stevinson, M Barsbey, T Birdal, and P Mediano. Correlations in the data lead to semantically rich feature geometry under superposition. *Mechanistic Interpretability Workshop at NeurIPS 2025*, 2025a.

-
- Lucas Prieto, Melih Barsbey, Pedro A. M. Mediano, and Tolga Birdal. Grokking at the edge of numerical stability. In *The Thirteenth International Conference on Learning Representations, ICLR, 2025b*.
- Aditi Raghunathan, Jacob Steinhardt, and Percy Liang. Certified defenses against adversarial examples. In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings*. OpenReview.net, 2018.
- Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Lin, Fred A. Hamprecht, Yoshua Bengio, and Aaron C. Courville. On the spectral bias of neural networks. In *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA*, volume 97 of *Proceedings of Machine Learning Research*, pp. 5301–5310. PMLR, 2019.
- Shashata Sawmya, Linghao Kong, Ilia Markov, Dan Alistarh, and Nir N Shavit. Wasserstein distances, neuronal entanglement, and sparsity. In *The Thirteenth International Conference on Learning Representations, 2025*.
- Adam Scherlis, Kshitij Sachan, Adam S Jermyn, Joe Benton, and Buck Shlegeris. Polysemanticity and capacity in neural networks. *arXiv preprint arXiv:2210.01892*, 2022.
- Ludwig Schmidt, Shibani Santurkar, Dimitris Tsipras, Kunal Talwar, and Aleksander Madry. Adversarially robust generalization requires more data. In *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems 2018, NeurIPS 2018, December 3-8, 2018, Montréal, Canada*, pp. 5019–5031, 2018.
- C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus. Intriguing properties of neural networks. In *2nd International Conference on Learning Representations, ICLR 2014, 2014*.
- Tomasz Tkocz. An upper bound for spherical caps. *The American Mathematical Monthly*, 119(7): 606–607, 2012.
- Donghua Wang, Wen Yao, Tingsong Jiang, Xiaohu Zheng, Junqi Wu, and Xiaoqian Chen. Improving the Transferability of Adversarial Examples by Feature Augmentation. *arXiv*, 2024.
- Christopher Wiedeman and Ge Wang. Disrupting adversarial transferability in deep neural networks. *Patterns*, 3(5), 2022.
- Eric Wong and Zico Kolter. Provable defenses against adversarial examples via the convex outer adversarial polytope. In *International conference on machine learning*. PMLR, 2018.
- Huan Zhang, Hongge Chen, Chaowei Xiao, Sven Gowal, Robert Stanforth, Bo Li, Duane S. Boning, and Cho-Jui Hsieh. Towards stable and efficient training of verifiably robust neural networks. In *8th International Conference on Learning Representations, ICLR, 2020*.
- Shufei Zhang, Zhuang Qian, Kaizhu Huang, Qiufeng Wang, Rui Zhang, and Xinping Yi. Towards better robust generalization with shift consistency regularization. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pp. 12524–12534. PMLR, 2021.

APPENDIX

A GENERATION OF ADVERSARIAL EXAMPLES

We generate AExs with PGD, the standard iterative attack method (Madry et al., 2018). An untar-
geted attack maximises the classifier’s loss $\mathcal{L}(f(\mathbf{x}'), y_{\text{true}})$, whereas a targeted attack minimises
 $\mathcal{L}(f(\mathbf{x}'), y_{\text{target}})$; both proceed via the update rule:

$$\mathbf{x}'^{(k+1)} = \Pi_S \left(\mathbf{x}'^{(k)} \pm \alpha \mathbf{g}_k \right) \quad (1)$$

where $+$ selects maximisation and $-$ minimisation. Here $\mathbf{x}'^{(0)}$ is an initial perturbed input, α the step
size, and $\mathbf{g}_k$ the (normalised) gradient $\nabla_{\mathbf{x}} \mathcal{L}(f(\mathbf{x}'^{(k)}), y_{\text{class}})$ with y_{class} equal to y_{true} or y_{target} ; the
operator Π_S projects onto the ϵ -ball $S = \{\mathbf{x}' \mid \|\mathbf{x}' - \mathbf{x}\|_p \leq \epsilon\}$ and clips to the input domain. We
instantiate two ℓ_p norms:

1. ℓ_∞ : Constraint $\|\delta\|_{\ell_\infty} = \max_i |\delta_i| \leq \epsilon$. $\mathbf{g}_k$ is $\text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\cdot))$. Π_S clips each $\mathbf{x}'_i$ to $[\mathbf{x}_i - \epsilon, \mathbf{x}_i + \epsilon]$.
2. ℓ_2 : Constraint $\|\delta\|_{\ell_2} = \sqrt{\sum_i \delta_i^2} \leq \epsilon$. $\mathbf{g}_k$ is $\nabla_{\mathbf{x}} \mathcal{L}(\cdot) / \|\nabla_{\mathbf{x}} \mathcal{L}(\cdot)\|_{\ell_2}$. Π_S rescales $\delta = \mathbf{x}' - \mathbf{x}$ if
 $\|\delta\|_{\ell_2} > \epsilon$ via $\delta \leftarrow \epsilon \cdot \delta / \|\delta\|_{\ell_2}$.

B OMITTED PROOFS & THEORETICAL RESULTS

To characterise how adversarial attacks exploit feature representations, we prove that the optimal
perturbation weights each input dimension by the alignment of its corresponding feature with the path
to the decision boundary. The analysis concerns our linear models, taken without ReLU activations,
and makes the interference mechanism of the main text fully explicit.

Consider an input $\mathbf{x} \in \mathbb{R}^d$ encoded via $\phi(\mathbf{x}) = \mathbf{W}_e \mathbf{x}$ into the latent representation $\mathbf{h} \in \mathbb{R}^m$, where
the columns of $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ form an overcomplete dictionary $\{\mathbf{v}_i\}_{i=1}^d$ ($m < d$). For the argmax
task, the trained encoder–decoder pair satisfies, empirically, $\mathbf{W}_d = \mathbf{W}_e^\top$, so that the logit for class
 j is $z_j = \mathbf{v}_j^\top \mathbf{h}$, with $\mathbf{v}_j$ the j -th column of $\mathbf{W}_e$, and the prediction is $\hat{y} = \arg \max_j z_j$. An input
perturbation maps to a latent perturbation through $\Delta \mathbf{h} = \mathbf{W}_e \delta$, where $\delta = (\delta_1, \dots, \delta_d)^\top$ collects
the perturbation coefficients.

The decision boundary between classes j and k is the locus of equal logits, $\mathcal{B}_{jk} = \{\mathbf{h} : z_j = z_k\}$,
equivalently $\{\mathbf{h} : (\mathbf{v}_k - \mathbf{v}_j)^\top \mathbf{h} = 0\}$. The vector $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is the normal to this pairwise
boundary, pointing in the direction that maximally increases the margin $z_k - z_j$. We restate the two
propositions before proving them.

Proposition 1 (Optimal targeted attack). The optimal input perturbations δ that maximise move-
ment from class j toward class k under constraint $\|\delta\|_2 = \epsilon$ satisfy:

$$\delta \propto \mathbf{W}_e^\top \mathbf{n}$$

Proof. To move a sample from being classified as j to being classified as k , we need to maximise the
logit margin $z_k - z_j$. Under perturbation δ , the new margin becomes:

$$z'_k - z'_j = \mathbf{v}_k^\top (\mathbf{h} + \Delta \mathbf{h}) - \mathbf{v}_j^\top (\mathbf{h} + \Delta \mathbf{h}) \quad (2)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^\top (\mathbf{h} + \mathbf{W}_e \delta) \quad (3)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^\top \mathbf{h} + (\mathbf{v}_k - \mathbf{v}_j)^\top \mathbf{W}_e \delta \quad (4)$$

The change in margin is $(\mathbf{v}_k - \mathbf{v}_j)^\top \mathbf{W}_e \delta = \delta^\top \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$. We seek:

$$\max_{\delta} \delta^\top \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j) \quad \text{subject to} \quad \|\delta\|_2 = \epsilon$$

Let $\mathbf{g} = \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$. By the Cauchy-Schwarz inequality:

$$|\delta^\top \mathbf{g}| \leq \|\delta\|_2 \|\mathbf{g}\|_2 = \epsilon \|\mathbf{g}\|_2$$

Equality is achieved when δ and $\mathbf{g}$ are parallel. Given the constraint $\|\delta\|_2 = \epsilon$:

$$\delta = \frac{\epsilon}{\|\mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)\|_2} \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

□

Corollary (Interference drives vulnerability). For a targeted attack from class j to class k , the adversarial perturbation magnitude for input feature i is:

$$|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$$

where $\mathbf{v}_i$ is the i -th column of $\mathbf{W}_e$. This quantity represents the differential interference between feature i and the class representations.

This corollary exposes the precise mechanism by which superposition creates adversarial vulnerability. Each input feature i is perturbed in proportion to $\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)$, its differential interference with the class representations. Under superposition, non-orthogonality guarantees that even semantically unrelated features possess non-zero inner products with $(\mathbf{v}_k - \mathbf{v}_j)$, and these residual inner products constitute exactly the exploitable interference. The multi-class setting compounds the effect: with k classes there are many pairwise boundaries, each contributing a distinct interference pattern. Adversarial perturbations act through these cross-feature dependencies, manipulating features that influence the decision solely through their interference with the class representations. This accounts for the empirical vulnerability directly—attacks succeed not by increasing the target-class features, but by exploiting the web of interference that superposition induces.

Prop. 1 characterises the optimal direction for moving from class j to k , *i.e.* the gradient that maximises the margin $z_k - z_j$. When an intervening class exists (with $z_j < z_i < z_k$ for some i), following this gradient may lead the model to predict i before reaching k ; iterative methods such as PGD accommodate this by recomputing the gradient at each step, so that the global attack path emerges from a sequence of local decisions rather than from a single optimisation. Moreover, although the analysis assumes $\mathbf{W}_d = \mathbf{W}_e^\top$ in accordance with our empirical observations, the result extends to arbitrary decoders: for a general decoder $\mathbf{W}_d \in \mathbb{R}^{k \times m}$ with rows $\mathbf{w}_j^\top$, the optimal perturbation becomes $\delta \propto \mathbf{W}_e^\top (\mathbf{w}_k - \mathbf{w}_j)$, which captures the interference between the encoder directions and the decoder weights.

Proposition 2 (Attack transferability). Consider encoders ϕ and ψ with basis matrices $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ and $\mathbf{W}'_e \in \mathbb{R}^{m \times d}$ whose columns are related by orthogonal transformation $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$). If both models use their encoder transpose as decoder (*i.e.*, $\mathbf{W}_d = \mathbf{W}_e^\top$ and $\mathbf{W}'_d = (\mathbf{W}'_e)^\top$), then both models have identical optimal input perturbation vectors for any targeted attack from class j to class k .

Proof. For model ϕ , the optimal perturbation from Prop. 1 is:

$$\delta^\phi \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

For model ψ with transformed columns $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$:

$$\delta^\psi \propto (\mathbf{W}'_e)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = (\mathbf{W}'_e)^\top (\mathbf{Q}\mathbf{v}_k - \mathbf{Q}\mathbf{v}_j)$$

Since $\mathbf{W}'_e = \mathbf{Q}\mathbf{W}_e$ (all columns are transformed), we have:

$$\delta^\psi \propto (\mathbf{Q}\mathbf{W}_e)^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{5}$$

$$= \mathbf{W}_e^\top \mathbf{Q}^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{6}$$

$$= \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j) \tag{7}$$

where the last equality uses $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. Thus δ^ϕ and δ^ψ have identical proportionality. Under the same norm constraint $\|\delta\|_2 = \epsilon$, we have $\delta^\phi = \delta^\psi$. □

Prop. 2 explains why attacks transfer between models trained under similar regimes. When two models learn feature representations that differ only by an orthogonal transformation—the same

geometric structure in a different orientation—they share identical vulnerability patterns in input space, because an orthogonal transformation preserves every inner product between features, and hence the entire interference structure that attacks exploit. This furnishes a direct explanation of our empirical finding that models trained on data with matching correlations transfer attacks at high rates: such models discover the same feature geometry up to rotation, and therefore inherit precisely the same adversarial vulnerabilities.

C TOY MODEL EXPERIMENTS

This section supplies additional details, extended results, and further discussion for the toy-model experiments of the main paper. We report model accuracies across a substantially wider range of parameters than the main text admits, clarifying how model capacity and data characteristics such as sparsity govern the learning process, and delineating the conditions under which feature superposition emerges. We then present further visual examples, complementing Fig. 1, that illustrate the mechanics of adversarial attacks under varied conditions.

C.1 HYPOTHESES TESTING FRAMEWORK

We now state explicitly the hypotheses underlying the three research questions of Sec. 3.

Research Q1: Do adversarial perturbations exploit superposition geometry?

- H_0 : Adversarial perturbations are random with respect to feature geometry.
- H_1 : Adversarial perturbations systematically exploit geometric relationships between superposed representations.

Research Q2: Do data correlations determine superposition geometry?

- H_0 : Input correlations have no systematic effect on learned geometries.
- H_1 : Input correlations determine geometric arrangements across model initializations.

Research Q3: Does shared geometry explain attack transferability?

- H_0 : Attack transferability is independent of geometric similarity.
- H_1 : Transferability increases with shared latent structure.

We test these hypotheses through the following controlled experiments:

- $H_1(1)$: We measure the input perturbation profile alignment with a class’s latent representation and the latent attack vector.
- $H_1(2)$: We systematically vary input correlations and measure resulting geometries.
- $H_1(3)$: We quantify transferability rates across models with varying geometric similarity.

C.2 ACCURACY OF TOY MODEL FOR A RANGE OF PARAMETERS

For conceptual clarity, the main paper reports the toy model chiefly in low-dimensional settings. To characterise its behaviour more broadly, this subsection details the classification accuracy of the CE toy model across varying hidden-layer size (h), number of classes (k), number of features, and sparsity level (S), thereby delineating the conditions under which the model comes to represent features in superposition. The sparsity level S is the probability that an individual input feature $x_i^{(j)}$ is set to zero, so that larger S denotes greater input sparsity; we report results as a function of the feature density $1 - S$, the complementary probability that a feature is non-zero. Together, Tab. 2 and Tab. 3 locate both the model’s performance limits and its capacity to learn latent representations in superposition.

C.3 CORRELATIONS CONSTRAIN ARRANGEMENTS

Tab. 4 extends the experiments of *Do Input Correlations Determine Latent Feature Geometry?* to a broader range of classes (k) and hidden dimensions (m), and confirms that the monotonic ordering—uncorrelated < paired < global—holds across every configuration tested.

Table 2: Classification accuracy of the CE toy model at a fixed bottleneck dimension ($m = 2$) across numbers of classes (k), total input features (features = $k \times 3$), and input feature densities ($1 - S$). The results show how input sparsity governs the rate of performance degradation as more classes must be superposed within a constrained latent space: at any fixed k , accuracy rises monotonically as the input becomes sparser.

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
3	9	2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
4	12	2	0.67	0.71	0.82	0.92	0.98	0.99	1.00	1.00
5	15	2	0.53	0.50	0.65	0.77	0.89	0.95	0.98	0.99
6	18	2	0.40	0.43	0.51	0.66	0.82	0.93	0.97	0.99
7	21	2	0.34	0.34	0.40	0.53	0.73	0.87	0.95	0.98
8	24	2	0.30	0.30	0.33	0.40	0.63	0.82	0.93	0.97
9	27	2	0.24	0.26	0.30	0.35	0.57	0.75	0.89	0.96
10	30	2	0.22	0.24	0.26	0.31	0.50	0.72	0.87	0.95
15	45	2	0.14	0.15	0.16	0.17	0.25	0.40	0.65	0.86
20	60	2	0.10	0.10	0.11	0.13	0.15	0.24	0.44	0.76
25	75	2	0.07	0.08	0.09	0.10	0.12	0.16	0.26	0.62
30	90	2	0.06	0.07	0.07	0.07	0.09	0.11	0.21	0.45

Table 3: Classification accuracy of the CE toy model across numbers of classes (k), total input features, bottleneck dimensions (m), and input feature sparsity levels (S), reported for varying numbers of features per class (p). These larger-scale configurations confirm that the sparsity-driven accuracy trend of Tab. 2 persists at higher dimensionality.

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
30	30	30	0.23	0.24	0.38	0.62	0.83	0.94	0.99	1.00
30	90	90	0.27	0.27	0.33	0.41	0.51	0.67	0.85	0.94
40	40	30	0.67	0.54	0.73	0.77	0.88	0.96	0.99	0.99
40	120	30	0.71	0.64	0.65	0.72	0.73	0.79	0.89	0.96
60	60	10	0.05	0.07	0.12	0.25	0.47	0.73	0.90	0.97
60	180	10	0.08	0.10	0.13	0.17	0.22	0.32	0.52	0.75
80	80	30	0.15	0.17	0.23	0.41	0.63	0.79	0.91	0.98
80	240	30	0.23	0.22	0.31	0.41	0.48	0.53	0.66	0.80
100	100	10	0.03	0.04	0.05	0.10	0.21	0.43	0.69	0.87
100	300	10	0.04	0.05	0.06	0.09	0.12	0.15	0.25	0.45

Table 4: Geometric similarity across correlation types and superposition configurations. In every (k, m) setting, geometric similarity increases monotonically from uncorrelated to paired to global correlations, matching the main-text result.

Correlation Type	k	m	Geometric Similarity
Uncorrelated	6	2	0.18 ± 0.15
	30	10	0.17 ± 0.02
	90	30	0.17 ± 0.01
Paired	6	2	0.47 ± 0.07
	30	10	0.26 ± 0.07
	90	30	0.25 ± 0.01
Global	6	2	0.92 ± 0.04
	30	10	0.88 ± 0.01
	90	30	0.80 ± 0.01

C.4 HOW THE TOY MODEL MAPS TO REAL-WORLD MODELS

How does scaling affect adversarial vulnerability? Empirically, larger models tend to be somewhat more robust, but the effect is weak; adversarial training yields clearer scaling trends, yet its gains remain largely specific to the attack type seen during training rather than conferring general robustness (Howe et al., 2025).

Two competing phenomena operate as models scale. Larger models command more capacity with which to represent concepts, yet general tasks such as next-token prediction over internet text present a long tail of useful concepts to capture, so that superposition remains prevalent even in frontier models (Lindsey et al., 2025). Dictionary-learning methods corroborate this: SAEs require ever-larger dictionaries for larger models (Gao et al., 2025), indicating that the number of represented features itself scales with model size. The fundamental tension that drives superposition—many features competing for a limited number of dimensions—therefore does not dissolve with scale.

Because both superposition and adversarial vulnerability persist at scale, the mechanism we characterise remains a relevant lens across model sizes. We emphasise, however, that persistence of superposition at scale does not by itself establish that the specific interference mechanism dominates in naturally trained large models; adjudicating that question requires *detecting* the geometry directly, which motivates the SAE-based programme of App. F. Characterising how the geometry of superposition evolves with scale, and whether that evolution can be steered to mitigate vulnerability, is a compelling direction for future work.

C.4.1 SEPARATING SUPERPOSITION EFFECTS FROM CAPACITY REDUCTION

Because increasing k/m simultaneously raises superposition pressure and, whenever m shrinks, reduces capacity, the two effects must be disentangled if superposition is to be credited as the cause. We achieve this by holding the bottleneck dimension fixed at $m = 2$ and varying only the number of classes k , which isolates superposition pressure from capacity. Tab. 5 reports model accuracy, robust accuracy, and transferability under this control, and the pattern is unambiguous: increased superposition pressure (k/m), at fixed capacity, is what drives the adversarial vulnerabilities we observe—not reduced dimensionality per se.

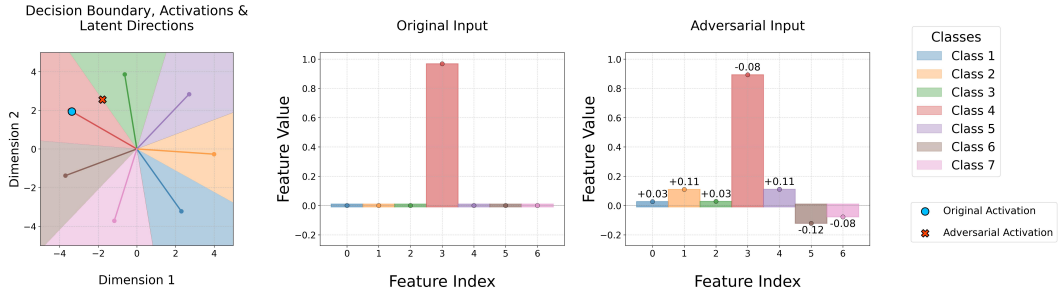
C.4.2 ADDITIONAL EXAMPLES OF AEXS IN TOY MODEL

Fig. 1 demonstrates how adversarial attacks exploit the interference between latent features in superposition. We reinforce that intuition with further visual examples (Fig. 5), which extend the main text by showcasing the following.

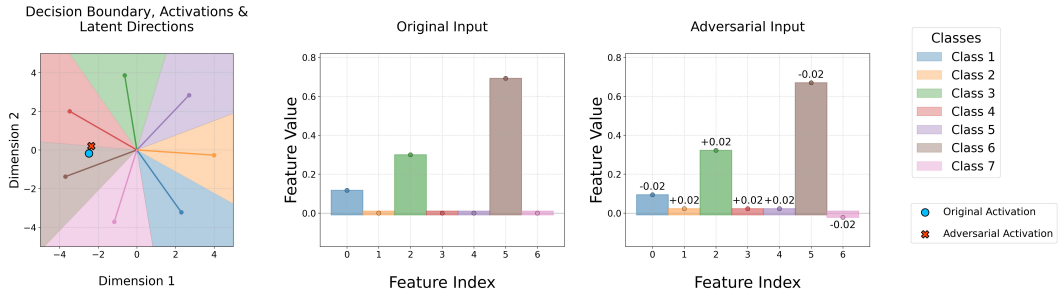
- Fig. 5a presents a further instance of the setup in Sec. 3.1 ($m = 2, k = 7$) under an ℓ_2 -norm PGD attack, exhibiting the IPP and the latent-space manipulations that produce misclassification.

Table 5: Performance metrics across correlation types, perturbation budgets (ε), and numbers of classes (k), with the bottleneck fixed at $m = 2$. At fixed capacity, robust accuracy falls and transferability rises as k (hence superposition pressure) grows, isolating superposition as the driver of vulnerability.

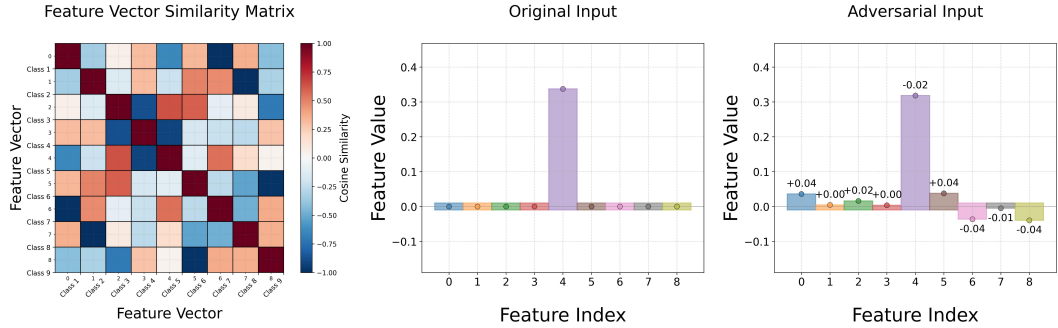
Correlation Type	Parameters		Performance (%)		
	ε	k	Accuracy	Robust Acc.	Attack Trans.
Uncorrelated	0.05	2	100.0	100.0	0.0
		4	100.0	94.9	21.2
		6	99.9	87.0	9.8
		8	99.8	77.7	17.5
	0.1	2	100.0	100.0	0.0
		4	100.0	88.1	9.8
		6	99.7	75.4	9.9
		8	99.6	56.3	23.5
	0.5	2	100.0	100.0	0.0
		4	100.0	99.4	0.0
		6	99.9	62.9	30.5
		8	99.5	24.9	39.0
Fully Correlated	0.05	2	100.0	100.0	0.0
		4	100.0	95.8	98.4
		6	97.3	70.6	98.4
		8	93.0	57.4	97.2
	0.1	2	100.0	100.0	0.0
		4	100.0	88.7	98.5
		6	97.7	45.6	99.7
		8	94.2	34.8	98.6
	0.5	2	100.0	100.0	0.0
		4	100.0	90.0	100.0
		6	97.2	34.6	100.0
		8	93.0	32.0	100.0



(a) An ℓ_2 -norm attack that reclassifies an input of class 4 as class 3. Left: the original and adversarial activations in latent space, together with the representation directions. Right: the original and perturbed input feature values.



(b) An ℓ_∞ -norm attack that reclassifies an input of class 6 as class 4. Left: the original and adversarial activations in latent space relative to the class latent directions. Right: the original and perturbed input feature values.



(c) An ℓ_2 -norm adversarial attack in a 7-class setup with an increased bottleneck dimension $m = 3$. The leftmost plot shows the cosine-similarity matrix between pairs of class latent representations.

Figure 5: Visualisations of AExs in the toy model, supplementing Fig. 1 by illustrating the attack mechanism in activation and input space under varied conditions.

- Fig. 5b shows the same configuration ($m = 2$, $k = 7$) under an ℓ_∞ -norm PGD attack applied to a less sparse input.
- Fig. 5c shows an example at increased bottleneck dimensionality ($m = 3$, $k = 7$) with an accompanying ℓ_2 -norm PGD attack, additionally displaying the feature-vector similarity matrix used to compute geometric similarity.

D CIFAR-10 EXPERIMENTS

To establish that the principles observed in the toy models carry to a standard vision architecture, Sec. 4 introduces experiments on CIFAR-10 (Krizhevsky, 2009) in which an engineered bottleneck

places the class representations in superposition. This appendix details that setup and presents extended results.

D.1 ARCHITECTURE & TRAINING INFORMATION

The base ViT (Dosovitskiy et al., 2020) comprised 6 transformer layers, an embedding dimension $d = 512$, and 8 attention heads per layer. CIFAR-10 images (32×32 pixels) were tokenised into 4×4 patches, each transformer block’s Multilayer Perceptron (MLP) used a hidden dimension of 512, and learned positional embeddings were employed.

The bottleneck comprised a linear encoder followed by a linear decoder: the encoder projected the ViT backbone’s pre-classification activations into an m -dimensional latent space ($m \in \{2, 3, 5, 10\}$), and the decoder mapped these m -dimensional representations back to the $k = 10$ CIFAR-10 classes.

The base ViT was trained on CIFAR-10 for 250 epochs with the Adam optimiser (default PyTorch parameters) at a learning rate of 0.001, a batch size of 512, a cosine-annealing learning-rate schedule, CE loss, and dropout, across five random seeds so as to quantify variability. Its weights were then frozen, and the bottleneck was trained for a further 30 epochs at a learning rate of 0.001. Preprocessing applied RandomCrop with 4-pixel padding, a resize to the target resolution (32×32 by default), and RandomHorizontalFlip for data augmentation.

D.1.1 NORMALISED ROBUST ACCURACY ACROSS PERTURBATION MAGNITUDES

Fig. 4 (right) reports normalised robust accuracy as a function of bottleneck dimension at a single ϵ . This subsection broadens that result by tabulating normalised robust accuracy under PGD attacks of varying strength, for both ℓ_2 -norm (Tab. 7) and ℓ_∞ -norm (Tab. 6) attacks, thereby giving a comprehensive picture of how the degree of superposition and the attack strength jointly determine robustness.

AExs for these evaluations were generated with PGD over 100 iterations at a step size $\alpha = 0.01$, and robust accuracy was evaluated on 500 samples per configuration. We report the mean normalised robust accuracy together with its standard deviation across five random seeds.

Table 6: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for ViT models with bottleneck dimension m on CIFAR-10, under ℓ_∞ -norm PGD attacks of varying perturbation magnitude (ϵ); robust accuracy is normalised by each bottlenecked model’s clean accuracy. Robustness increases monotonically with m at every ϵ , corroborating Sec. 4 across the full range of attack strengths.

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.001	96.0% $\pm$ 0.4%	97.4% $\pm$ 0.7%	97.9% $\pm$ 0.4%	98.1% $\pm$ 0.3%
0.01	61.7% $\pm$ 3.6%	69.6% $\pm$ 3.5%	77.0% $\pm$ 0.9%	81.8% $\pm$ 3.2%
0.05	4.9% $\pm$ 1.5%	6.7% $\pm$ 1.6%	9.3% $\pm$ 0.6%	10.5% $\pm$ 0.8%
0.1	0.1% $\pm$ 0.2%	0.2% $\pm$ 0.3%	0.2% $\pm$ 0.4%	0.2% $\pm$ 0.4%
0.5	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%

D.1.2 ATTACK TRANSFERABILITY ACROSS PERTURBATION MAGNITUDES

We report attack transferability jointly across perturbation magnitudes (ϵ) and bottleneck dimensions (m): Tab. 9 gives the ℓ_2 -norm transferability and Tab. 8 the ℓ_∞ -norm transferability.

D.2 CORRELATIONS IN CLASS FEATURES

Whereas in Sec. 3 the superposition arrangement was driven by correlations among the inputs, the relevant correlations here are those among the representations at this point in the network rather

Table 7: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for ViT models with bottleneck dimension m on CIFAR-10, under ℓ_2 -norm PGD attacks of varying perturbation magnitude (ϵ); robust accuracy is normalised by each bottlenecked model’s clean accuracy. The results reinforce Sec. 4: robustness declines with smaller m (greater superposition) and with larger ϵ .

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.1	90.4% $\pm$ 1.7%	91.8% $\pm$ 0.7%	94.6% $\pm$ 1.2%	95.0% $\pm$ 0.7%
0.5	58.5% $\pm$ 5.0%	60.8% $\pm$ 5.0%	69.2% $\pm$ 1.8%	72.6% $\pm$ 2.8%
1.0	41.7% $\pm$ 4.8%	44.0% $\pm$ 3.0%	50.4% $\pm$ 0.6%	54.9% $\pm$ 1.7%
2.0	34.0% $\pm$ 4.6%	36.2% $\pm$ 3.3%	41.5% $\pm$ 1.3%	47.4% $\pm$ 2.4%
5.0	30.2% $\pm$ 4.1%	32.9% $\pm$ 3.5%	37.3% $\pm$ 1.9%	42.7% $\pm$ 1.0%

than among the input classes themselves. At the classification layer these correlations plausibly mirror the confusion matrix—*i.e.* the pattern of which classes are misclassified as which—so that the classes that cluster together are precisely those the network finds inherently similar and tends to confuse. We test this prediction by repeating the experiment while instead fine-tuning the base ViT from timm/vit_base_patch16_384, pre-trained on ImageNet-21k (14 million images, 21,843 classes) and ImageNet (1 million images, 1,000 classes). After fine-tuning on CIFAR-10 and applying the bottleneck procedure of Sec. 4, the resulting geometry exhibits random ordering across initialisations with approximately equal feature spacing (*i.e.* neural collapse Kothapalli (2023)): performance approaches 100%, the confusion matrix approaches the identity, and the superposed structure dissolves. Models trained solely on CIFAR-10, by contrast, converge to similar geometries because they share a common difficulty structure—the same class pairs remain hard to distinguish—which in turn yields consistent superposition patterns. This confirms that it is shared representational correlation, and not any property of the raw inputs, that drives the consistent geometry.

D.2.1 RESNET-18

To establish that the mechanism is not specific to a single architecture, we repeat the experiments with a ResNet-18 (He et al., 2016) base model in place of the ViT, attaining a slightly higher clean accuracy of 92% (against 89% for the ViT). Although robust accuracy falls marginally faster for the ResNet-18 at matched ϵ , the same qualitative signatures hold: normalised robust accuracy declines, and transferability rises, as bottleneck pressure increases. Under the same engineered-bottleneck intervention, the mechanism thus recurs across both convolutional and transformer backbones.

E MODULAR ADDITION

Sec. 5 introduces a modular-addition task that probes adversarial vulnerability in a regime where the features are orthogonal rather than superposed. This section provides the detailed experimental setup, additional results, and a fuller account of why we regard this style of analysis as broadly useful.

E.1 LEARNED ALGORITHMIC REPRESENTATION IN MODULAR ADDITION

As shown by Nanda et al. (2023) and Prieto et al. (2025b), the network discovers and implements a trigonometric algorithm for this task: the encoder projects inputs onto non-basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at the key frequencies $\omega_k = 2\pi k/P$. These features are represented orthogonally—not in superposition—and their sinusoidal values are recoverable by linear probes. The subsequent MLP applies sum-angle identities to compute $\cos((a+b)\omega_k)$ for several k , and these terms then interfere constructively at the logits to select the correct class. AExs are generated by relaxing the one-hot constraint and perturbing $\mathbf{x}_j \in \mathbb{R}^P$ (encoding integer j) to $\mathbf{x}' = \mathbf{x} + \delta$ via ℓ_2 - and ℓ_∞ -norm PGD, attacking only a while holding b fixed.

Following Nanda et al. (2023), the network learns a structured trigonometric algorithm in which W_e encodes the sine and cosine of each integer at the key frequencies as linearly decodable values.

Table 8: Attack transferability (%) for ℓ_∞ -norm PGD attacks on CIFAR-10 ViT models. Each entry reports transfer from a model trained with a given ‘Source Seed’ (e.g. Seed 10) to three target models trained with the seeds in the sub-header (e.g. ‘vs. Seeds 20/30/40’); the three slash-separated values give transferability to those three target seeds in order. All models in a row share the bottleneck dimension m , and the ‘Mean $\pm$ Std’ column averages over all 12 source–target pairings for each (ϵ, m) . The results substantiate the claim of Sec. 4 that greater superposition yields more consistent latent geometries and hence higher transferability.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.001	2	77.8/66.7/44.4	25.0/62.5/25.0	72.7/72.7/72.7	58.3/75.0/58.3	59.3 $\pm$ 17.7
	3	57.1/28.6/28.6	14.3/28.6/28.6	63.6/72.7/72.7	42.9/42.9/42.9	43.6 $\pm$ 18.4
	5	70.0/50.0/50.0	57.1/71.4/57.1	50.0/25.0/25.0	16.7/50.0/33.3	46.3 $\pm$ 16.9
	10	55.6/55.6/66.7	12.5/37.5/25.0	16.7/33.3/16.7	28.6/71.4/28.6	37.3 $\pm$ 19.3
0.01	2	60.3/52.6/48.7	63.8/42.0/60.9	53.8/40.9/52.7	51.2/55.8/46.5	52.4 $\pm$ 6.9
	3	42.9/46.4/50.0	46.4/42.9/49.1	44.2/38.9/44.2	42.9/45.1/41.8	44.6 $\pm$ 3.0
	5	43.3/41.1/43.3	39.2/30.4/41.8	46.0/40.2/47.1	41.2/35.3/32.9	40.2 $\pm$ 4.8
	10	42.5/46.0/44.8	38.0/39.4/46.5	43.5/47.8/47.8	32.6/40.0/35.8	42.1 $\pm$ 4.7
0.05	2	35.0/36.4/37.8	38.8/36.0/46.3	37.4/31.3/41.7	42.0/42.0/40.7	38.8 $\pm$ 3.8
	3	29.9/29.9/37.1	30.4/30.1/35.6	30.2/30.2/35.1	30.1/31.9/34.4	32.1 $\pm$ 2.6
	5	26.8/25.3/29.2	25.2/21.5/26.4	28.0/28.0/26.5	26.0/26.6/21.9	25.9 $\pm$ 2.2
	10	27.1/24.9/30.4	21.6/23.0/28.4	25.3/28.1/30.6	21.0/24.9/20.1	25.4 $\pm$ 3.4
0.1	2	34.2/35.6/38.2	36.7/34.9/44.5	37.9/30.0/40.8	41.2/42.0/40.3	38.0 $\pm$ 3.8
	3	29.6/28.9/35.2	28.6/29.2/34.8	29.1/28.2/34.0	28.8/31.1/32.8	30.9 $\pm$ 2.5
	5	25.4/23.3/27.0	23.8/19.6/24.6	25.9/26.7/25.4	24.6/24.3/20.7	24.3 $\pm$ 2.1
	10	24.8/22.6/29.7	20.1/21.6/26.1	22.5/25.5/28.0	19.5/23.3/18.7	23.5 $\pm$ 3.3

Consequently, scaling a feature vector alters the encoded sinusoidal value itself rather than merely its activation intensity—a distinction that proves central to the attack mechanism. The integer embeddings $\mathbf{W}_e^j$ project onto a Fourier-like basis, inducing a circular, periodic structure in embedding space; unlike the arrangements produced by data correlations, this circularity arises directly from the sine/cosine encoding of the (uncorrelated) input integers.

The model learns to employ approximately seven key frequencies (ω_k). Each embedding W_e^j for an input integer x_j predominantly encodes its fourteen corresponding trigonometric features, $\cos(\omega_i x_j)$ and $\sin(\omega_i x_j)$ for $i \in \{1, 2, \dots, 7\}$, so that the concatenated 200-dimensional vector supplies roughly 28 key trigonometric features (fourteen for a and fourteen for b) to the MLP. The MLP combines these through sum-angle identities (e.g. $\cos(A + B) = \cos A \cos B - \sin A \sin B$) to compute $(x_1 + x_2) \pmod{P}$. The algorithmic brittleness studied in Sec. 5 arises precisely from the adversarial perturbation of these learned trigonometric features.

E.2 EXPERIMENT DETAILS FOR MODULAR ADDITION TASK

The network is trained for addition modulo 113 on integer pairs (a, b) with $a, b \in \{0, \dots, 112\}$. Each integer is mapped to a 100-dimensional vector by a shared embedding $\mathbf{W}_E \in \mathbb{R}^{113 \times 100}$, and the embeddings $\mathbf{W}_E^a$ and $\mathbf{W}_E^b$ are concatenated into a 200-dimensional feature vector. This vector feeds a 3-hidden-layer MLP (200 ReLU neurons per layer) whose output layer produces 113 classification logits. The model is trained end-to-end with cross-entropy loss and AdamW (weight decay 4), reaching 100% test accuracy. AExs are generated with PGD under the ℓ_2 and ℓ_∞ norms at varying ϵ , perturbing the embedding $\mathbf{W}_E^a$ (with b held fixed) by adding δ so as to misclassify $(a + b) \pmod{113}$.

Table 9: Attack transferability (%) for ℓ_2 -norm PGD attacks on CIFAR-10 ViT models. The layout of source-to-target transferability follows Tab. 8 (see its caption for the full explanation). These ℓ_2 results further support the claim of Sec. 4 that greater superposition increases attack transferability.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.1	2	70.6/64.7/41.2	60.0/50.0/60.0	64.3/60.7/57.1	48.1/63.0/51.9	57.6 ± 8.0
	3	39.1/39.1/47.8	50.0/60.7/50.0	48.1/55.6/51.9	48.0/28.0/40.0	46.5 ± 8.4
	5	43.5/34.8/43.5	31.6/31.6/42.1	42.9/35.7/21.4	30.4/39.1/34.8	35.9 ± 6.4
	10	45.5/59.1/45.5	19.0/28.6/28.6	37.5/43.8/43.8	23.8/52.4/38.1	38.8 ± 11.4
0.5	2	59.6/52.1/48.9	60.8/35.4/60.8	54.1/39.4/51.4	43.8/49.5/46.7	50.2 ± 7.7
	3	38.7/40.6/48.1	36.1/36.8/41.7	37.4/36.5/43.5	34.4/34.4/36.7	38.7 ± 3.9
	5	38.4/32.0/37.6	29.4/25.7/32.1	39.1/33.9/38.3	31.8/29.9/32.7	33.4 ± 4.0
	10	38.3/37.4/37.4	26.5/28.4/31.4	32.4/35.3/35.3	23.4/28.2/25.0	31.6 ± 5.0
1.0	2	50.8/43.7/44.4	51.7/36.4/50.0	46.1/32.9/44.7	42.7/46.7/45.3	44.6 ± 5.3
	3	35.7/36.9/41.7	31.8/36.4/42.6	33.7/32.0/37.3	34.9/32.0/35.5	35.9 ± 3.3
	5	32.1/28.9/33.2	30.9/23.0/30.9	35.9/31.5/33.7	27.7/28.2/27.1	30.3 ± 3.3
	10	32.3/32.3/38.5	23.7/27.7/29.9	29.1/32.0/34.3	22.2/25.6/21.1	29.1 ± 5.0
2.0	2	47.0/41.6/44.3	48.2/36.2/48.2	43.9/33.5/42.1	43.4/43.4/42.8	42.9 ± 4.2
	3	31.9/37.7/40.3	31.1/35.6/40.6	32.6/31.1/36.8	35.4/33.3/34.3	35.1 ± 3.2
	5	29.9/25.4/32.1	26.3/22.1/26.7	32.3/31.8/29.5	27.9/27.9/23.6	28.0 ± 3.2
	10	30.8/30.4/35.7	22.5/24.9/28.6	24.8/29.7/32.7	20.4/26.1/19.0	27.1 ± 4.9
5.0	2	44.7/46.7/40.8	45.3/37.3/48.7	42.7/33.9/41.5	42.3/42.3/40.6	42.2 ± 3.8
	3	32.3/35.8/40.8	31.3/32.6/39.9	31.0/30.0/33.8	35.0/33.0/33.0	34.1 ± 3.2
	5	30.6/24.1/28.6	28.3/20.9/24.8	31.7/31.3/27.4	28.1/29.0/23.2	27.3 ± 3.3
	10	29.8/27.4/32.7	19.7/24.1/28.9	24.9/29.9/31.2	18.8/25.8/19.2	26.0 ± 4.6

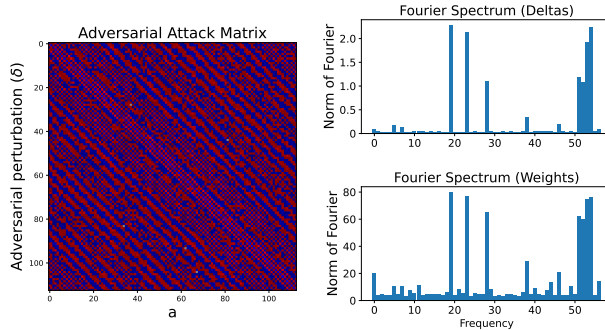


Figure 6: Left: each column is the mean attack δ added to a so as to successfully attack $a + b \bmod p$. Right: the norms of the Fourier components of δ and of $W_{MLP_{out}}$, demonstrating that the attack concentrates precisely on the model’s learned key frequencies.

Table 10: Minimum ℓ_2 -norm budget required to successfully attack $0 + 0 \bmod p$, by attack type. The informed attack, constructed purely from the learned frequencies, nearly matches gradient-based PGD (0.22 vs. 0.17) while far undercutting every naive baseline.

Attack Method	Success ϵ
PGD	0.17
Informed Attack	0.22
Single a	0.97
ℓ_∞ Noise	2.04
Uniform δ	6.20

E.3 ALGORITHMIC BRITTLINESS

The frequency-aligned perturbation induces misclassification by corrupting the sinusoidal values that the MLP reads off the encoder features. The hypothesised neuron-aligned case affords the clearest intuition: W_e contains, for each ω_k , a row whose entry in column w_e^a holds the value $\sin(\omega_k a)$

or $\cos(\omega_k a)$. For input a , the one-hot $\mathbf{x}_a$ selects $\mathbf{w}_e^a$, yielding $\mathbf{h}_a = \mathbf{w}_e^a$. The perturbation δ then converts the one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$, so that $\mathbf{h}'_a = \mathbf{W}_e \mathbf{x}'_a = (1 + \epsilon_a) \mathbf{h}_a + \sum_{k \neq a} \epsilon_k \mathbf{w}_e^k$. The contributions from the $k \neq a$ sinusoidal terms corrupt the sinusoidal values, disrupting the MLP's downstream computation and thereby producing misclassification. Because the algorithm depends on the constructive interference of every ω_k at the logits, perturbing a single ω_k is insufficient—the remaining components overwhelmingly support the correct answer—so that an effective δ must maximally alter several frequency components simultaneously.

E.4 ANALYTIC CONSTRUCTION OF ATTACKS AND BASELINE COMPARISONS

The main paper shows that a characterisation of the learned trigonometric algorithm enables the analytic construction of adversarial attacks, and Sec. 5 benchmarks this ‘informed attack’ against PGD and several naive baselines. Here we specify those baselines precisely.

- **PGD Attack:** standard PGD used to locate the adversary.
- **Informed Attack:** the analytically constructed perturbation, formed from combinations of cosines at the learned frequencies.
- **Single a :** the minimal perturbation of a single input entry that causes misclassification.
- **ℓ_∞ Noise:** random noise δ_{noise} added to input a , with each component drawn independently and uniformly from $[-\epsilon, \epsilon]$; we report the smallest ϵ that induces misclassification.
- **Uniform δ :** a perturbation with all components equal, $\delta_j = c \cdot 1$ for a scalar c ; we report the magnitude of c at which adding δ causes misclassification.

E.5 DETAILS ON CERTIFIED ROBUST TRAINING

To assess how defences bear on the identified vulnerability, we performed certified robust training aimed at rendering the network provably robust to ℓ_∞ -norm perturbations. We employed the abstraction-based technique RSIP-IBP (Zhang et al., 2020; Henriksen & Lomuscio, 2023), a certified-training method (Wong & Kolter, 2018; Raghunathan et al., 2018; Palma et al., 2022) that combines Interval Bound Propagation (IBP) (Katz et al., 2017; Ehlers, 2017) with Reversed Symbolic Interval Propagation (RSIP) to compute per-layer reachable output sets and thereby certify robustness within a given perturbation budget. The training loss combined a standard loss term with an RSIP term. Certified training ran for 1000 epochs (the first 200 constituting a warm-up phase) at a learning rate of 1×10^{-3} and weight decay 1×10^{-4} , using the same train/test splits as standard training. Tab. 11 summarises the results across training perturbation budgets (ϵ_{train}) and attack budgets (ϵ_{attack}).

Table 11: Verified robust accuracy (%) for modular-addition models under ℓ_∞ -norm perturbations, comparing a standard (pre-robustification) model against models certifiably trained with RSIP-IBP at various training budgets (ϵ_{train}), evaluated across attack budgets (ϵ_{attack}). Consistent with the discussion in Sec. 5, certified training improves robustness up to the certified ϵ_{train} , yet attacks still succeed at higher ϵ_{attack} by exploiting the same algorithmic sensitivities.

Attack Epsilon (ϵ_{attack})	Train Epsilon (ϵ_{train})	Standard Acc. (Pre-Rob., %)	Robust Acc. (Pre-Rob., %)	Standard Acc. (Robust, %)	Robust Acc. (Robust, %)
5.0×10^{-4}	10^{-4}	100.0	100.0	100.0	100.0
	10^{-3}				100.0
1.0×10^{-3}	10^{-4}	100.0	76.0	100.0	93.8
	10^{-3}				100.0
2.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				95.7
5.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				85.3

As noted in Sec. 5, subjecting these robustly trained models to PGD leaves the character of the successful perturbations unchanged: they continue to exploit the same key frequency components of the learned algorithm. Certified training merely raises the perturbation magnitude required

for misclassification at the epsilons against which the model was certified; it does not alter the attack’s alignment with the model’s algorithmic sensitivities. The fragility is displaced in budget, not removed—which is precisely why we argue for defences directed at the exploited semantics rather than at norm bounds alone.

E.6 ON A POTENTIAL RECONCILIATION WITH SUPERPOSITION VULNERABILITY

Although the vulnerability studied here arises in the *absence* of feature superposition, it bears a striking phenomenological resemblance to the superposition-driven case, in the form of a shared dependence on sparsity in the input representation. This motivates our *reconciliation conjecture*: algorithmic vulnerabilities may persist in networks with orthogonal representations whenever sparsity assumptions are violated upstream. If earlier layers emit noisy or non-sparse features—possibly as a consequence of adversarial interference—these deviations propagate into otherwise well-isolated downstream circuits, so that a circuit which operates correctly on clean inputs nonetheless fails because the input it receives is already effectively perturbed. This supplies a plausible route by which algorithm-disruption mechanisms such as the frequency attacks studied here could be triggered even in the absence of superposition. The interaction between upstream superposition and downstream algorithmic brittleness—and, more broadly, a layer-by-layer account of the specific vulnerability mechanisms that AExs exploit—remains underexplored, and we regard a detailed investigation grounded jointly in empirical evidence and theoretical modelling as a promising direction for future work.

F INSPECTING ADVERSARIAL PERTURBATIONS VIA SAE FEATURES

The analyses in the main paper concern settings with either explicitly defined class features (the toy model) or class-level representations placed in a bottleneck (the ViT experiments). In large, practical networks, by contrast, interference occurs among unlabelled latent features distributed across all layers. As the main-paper discussion of future work notes, SAEs offer a promising unsupervised route to extracting a larger set of linear features from NNs (Bricken et al., 2023). This appendix reports a preliminary investigation that uses SAE-extracted features to characterise how a large-scale ViT responds internally to AExs as opposed to clean inputs. We stress at the outset that these initial results are intended to establish the *feasibility* of the approach for the future feature-level study of adversarial phenomena in large models, and emphatically not to present a conclusive set of findings; they neither confirm nor refute the presence of the interference mechanism in naturally trained models. We first describe the setup and validate the behaviour of the SAEs on adversarial inputs, and then present exploratory layer-wise feature-difference analyses.

F.1 EXPERIMENT SETUP

We analyse SAE-extracted features from the vision encoder of the LAION/CLIP-ViT-B-32-DataComp.XL-s13B-b90K model, a 12-layer ViT attaining $\approx 70\%$ zero-shot accuracy on ImageNet-1k Deng et al. (2009). The open-source Prisma SAEs map the 768-dimensional activation space to a dictionary of 49,152 features (a $64\times$ expansion) (Joseph et al., 2025). We examine SAEs trained on either mean patch-token or CLS-token representations, across all layers, and analyse features activating above a high threshold (0.1, which captures the most strongly activated features) and a low threshold (0.001). For the SAE feature comparisons we generate 10,000 original/successfully-attacked image pairs (ℓ_∞ -norm, $\epsilon = 0.02$). This setup supports an initial exploration of how adversarial perturbations manifest at the level of SAE features.

F.2 SAE VALIDITY FOR ADVERSARIAL INPUTS

Because AExs are out-of-distribution inputs, and we intend to interpret them with SAEs trained in the standard manner, we first perform a basic validation. This subsection measures the effect on model accuracy and SAE reconstruction of inserting an SAE at inference time for clean and adversarial inputs, so as to establish—before we interpret any differences in feature activations—whether the SAEs continue to process features relevant to classification even under adversarial conditions.

Table 12: Impact on ImageNet-1k classification accuracy of inserting a pre-trained SAE (one layer at a time) at inference, for original, ℓ_∞ adversarial ($\epsilon = 0.02$), and random noisy inputs. ‘Accuracy’ is the ViT’s top-1 accuracy; ‘Original input + SAE’ denotes passing the clean ViT activations at a given layer through the SAE and forwarding its reconstruction to the next layer. The slight ‘de-attacking’ effect—improved accuracy for ‘Adv. input + SAE’ over ‘Adv. input’—indicates that the SAEs filter some adversarial noise, yet the model still overwhelmingly misclassifies, confirming that the SAEs preserve the salient adversarial changes responsible for misclassification.

Metric	Layer 2	Layer 5	Layer 8	Layer 11
Accuracy				
Original input	1.00	1.00	1.00	1.00
Original input + SAE	0.99	0.99	0.99	0.86
Adv. input	0.00	0.00	0.00	0.00
Adv. input + SAE	0.02	0.01	0.01	0.11
Noisy input	0.98	0.98	0.98	0.98
Noisy input + SAE	0.98	0.98	0.98	0.85
SAE Improvement				
Original input	-0.01	-0.01	-0.01	-0.14
Adv. input	0.02	0.01	0.01	0.11
Noisy input	-0.00	-0.00	-0.00	-0.13

F.3 LAYER-WISE SAE FEATURE DIFFERENCES

This subsection presents initial, exploratory visualisations comparing SAE feature activations across ViT layers for original, ℓ_∞ adversarial, and random noisy inputs. Their purpose is to identify avenues for future, more detailed investigation of how adversarial attacks distinctively alter sparse feature representations. We examine the following quantities.

- Activation count: histograms comparing the number of SAE features activated above a set threshold in a layer, across original, adversarial, and noisy images, probing whether attacks characteristically alter feature sparsity.
- Feature overlap: the number of commonly activated features between the (original vs. adversarial) and (original vs. noisy) conditions, probing whether attacks activate a distinctly different feature set.
- Jaccard similarity: the Jaccard similarity between the sets of activated features, a normalised measure of overlap.
- Mean activation: histograms of the mean activation of features active above a threshold, probing changes in the intensity of the (key) features.
- Distinct features: the number of unique features activated across an image set in each condition.

F.4 CLS TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection presents a preliminary analysis of SAE features derived from CLS tokens, restricted to those features activated above the 0.1 threshold. Figures 7 through 10 visualise, respectively, the activation-count histogram, feature overlap, Jaccard similarity, and mean-activation histogram, comparing original, adversarial, and random noisy inputs across the ViT layers.

F.5 CLS TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Here we examine CLS-token SAE features at the lower activation threshold of 0.001, so as to capture more than only the most intensely activated features. The figures show the activation-count histogram (Fig. 11), feature overlap (Fig. 12), Jaccard similarity (Fig. 13), and mean-activation histogram (Fig. 14) across layers for original, adversarial, and noisy inputs.

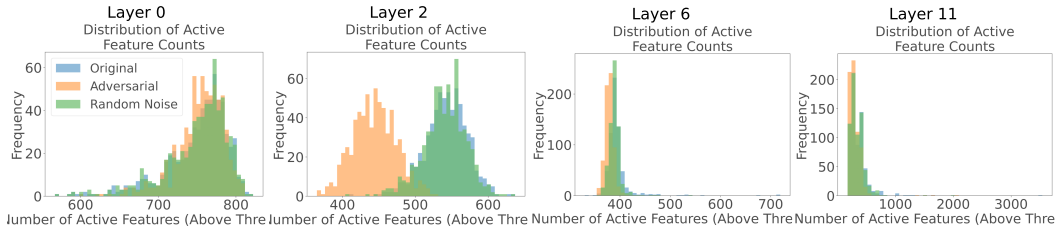


Figure 7: Histogram of activation counts for CLS token SAE features (threshold 0.1).

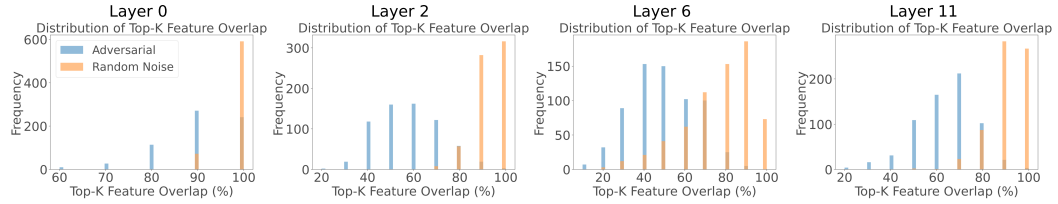


Figure 8: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features (threshold 0.1).

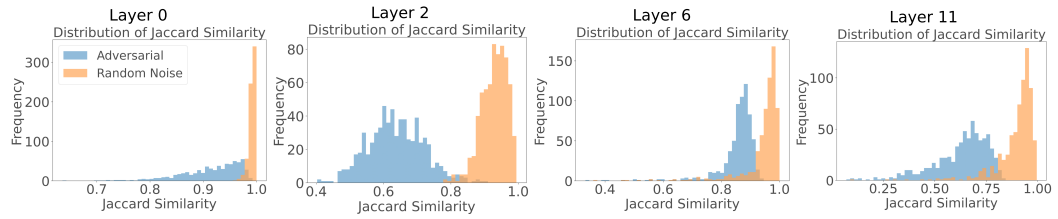


Figure 9: Comparison of Jaccard similarity for CLS token SAE features (threshold 0.1).

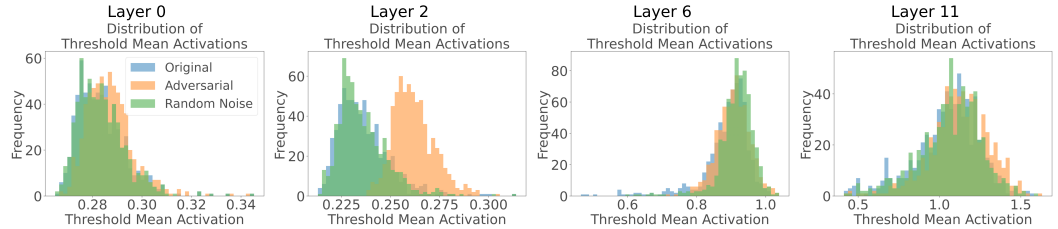


Figure 10: Histogram of mean thresholds for CLS token SAE features (threshold 0.1).

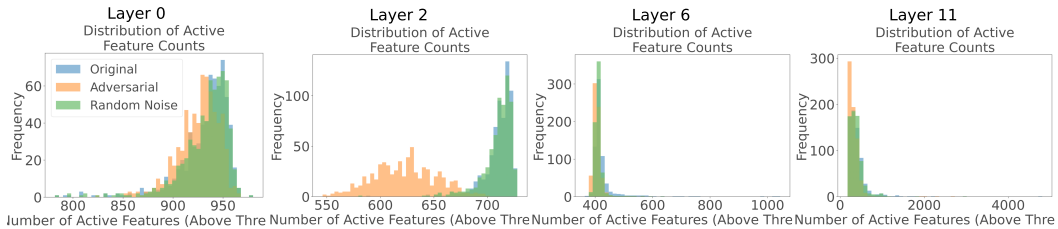


Figure 11: Histogram of activation counts for CLS token SAE features (threshold 0.001), with noise.

F.6 PATCH TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection turns to SAE features derived from the mean of the patch tokens, again at the high activation threshold of 0.1. We compare activation-count histograms (Fig. 15), distinct-feature-count

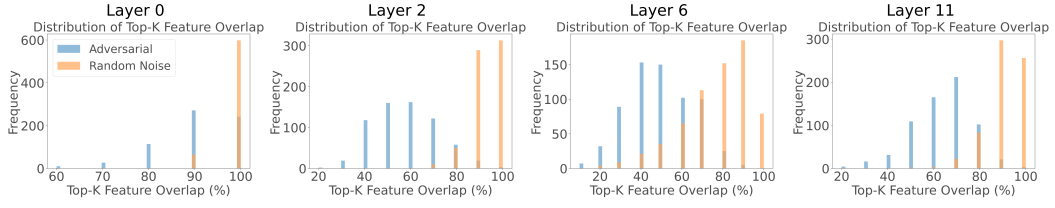


Figure 12: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features (threshold 0.001).

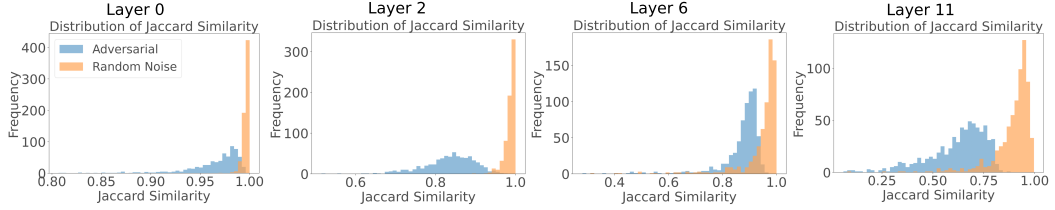


Figure 13: Comparison of Jaccard similarity for CLS token SAE features (threshold 0.001).

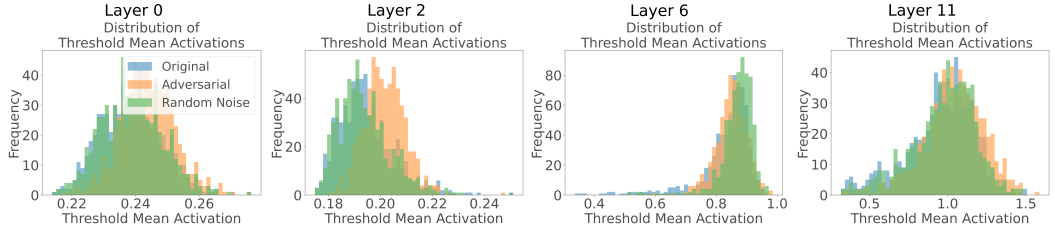


Figure 14: Histogram of mean activations for CLS token SAE features (threshold 0.001), with noise.

histograms (Fig. 16), feature overlap (Fig. 17), Jaccard similarity (Fig. 18), and mean-activation histograms (Fig. 19) across original, adversarial, and noisy inputs.

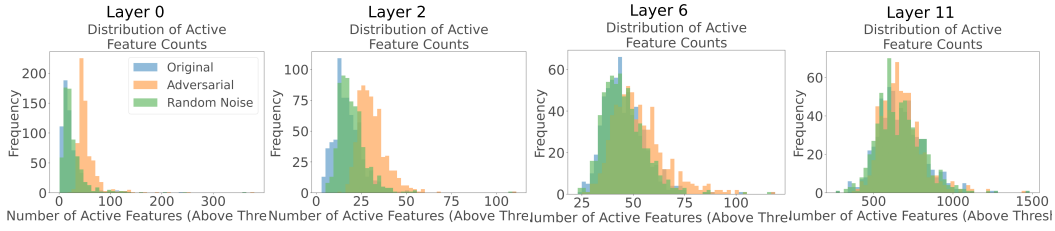


Figure 15: Histogram of activation counts for patch token SAE features (threshold 0.1), with noise.

F.7 PATCH TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Finally, we explore SAE features from the patch-token means at the low activation threshold of 0.001. The figures show the activation-count histograms (Fig. 20), distinct-feature counts (Fig. 21), feature overlap (Fig. 22), Jaccard similarity (Fig. 23), and mean-activation histograms (Fig. 24) across original, adversarial, and noisy inputs.

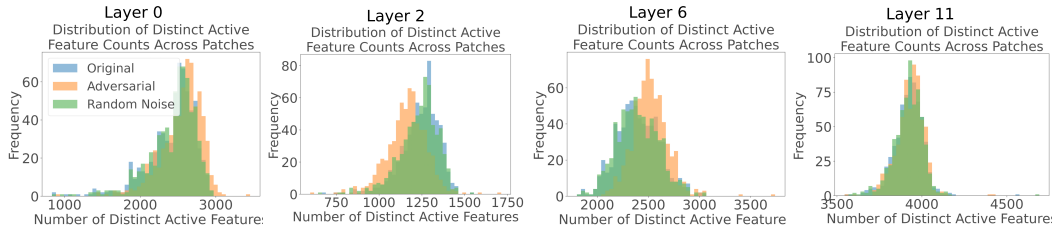


Figure 16: Histogram of distinct patch features for patch token SAE features (threshold 0.1), with noise.

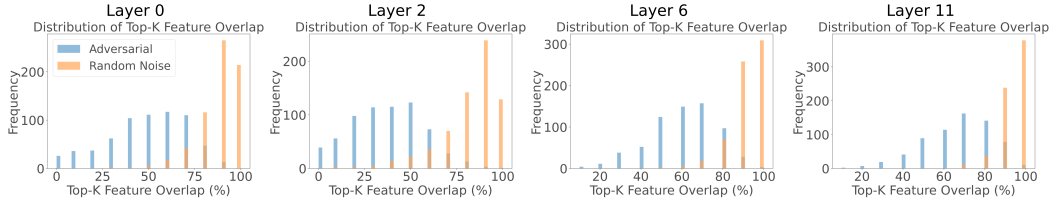


Figure 17: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features (threshold 0.1).

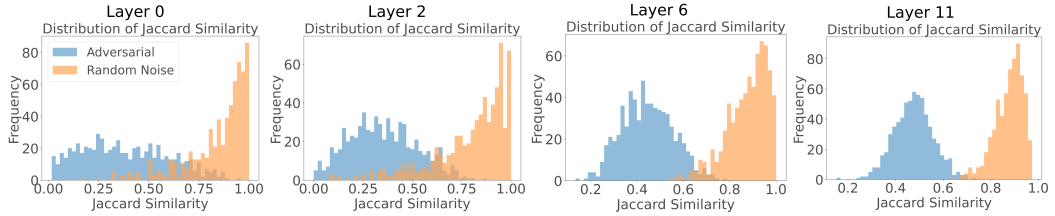


Figure 18: Comparison of Jaccard similarity for patch token SAE features (threshold 0.1).

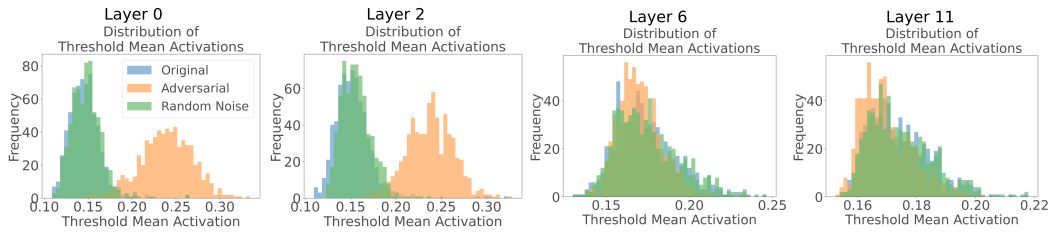


Figure 19: Histogram of mean thresholds for patch token SAE features (threshold 0.1), with noise.

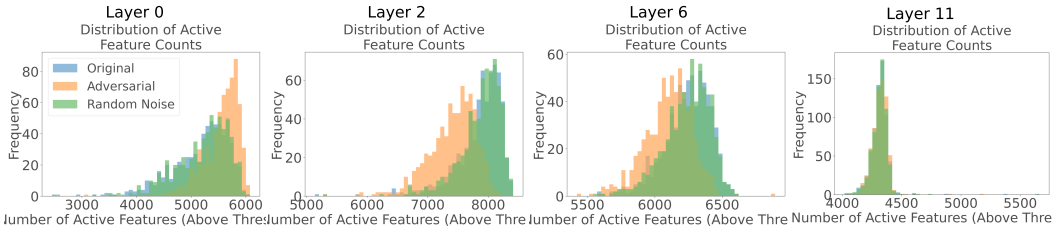


Figure 20: Histogram of activation counts for patch token SAE features (threshold 0.001), with noise.

G ACRONYMS

AEx adversarial example

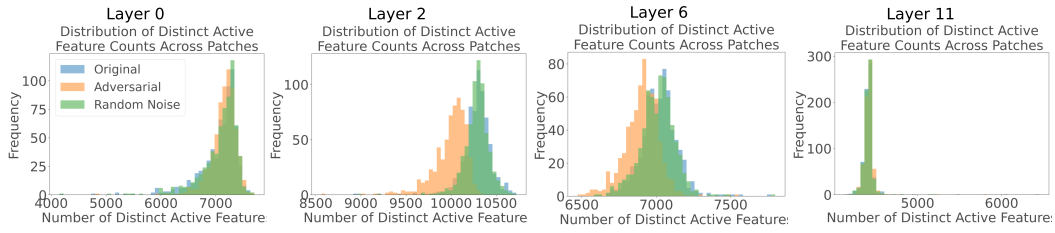


Figure 21: Histogram of distinct patch features for patch token SAE features (threshold 0.001), with noise.

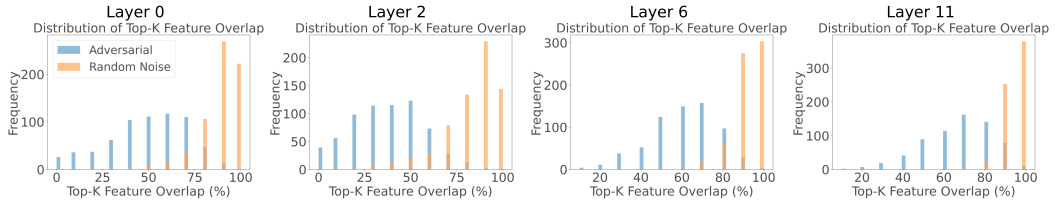


Figure 22: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features (threshold 0.001).

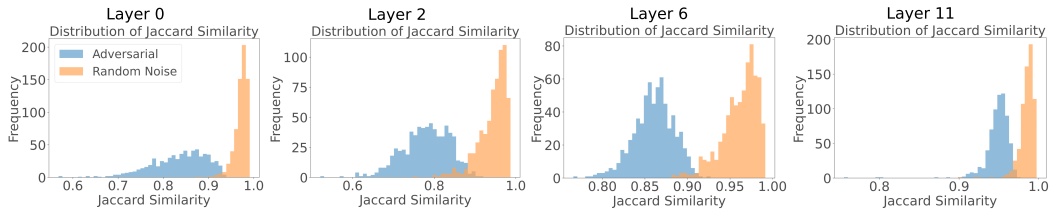


Figure 23: Comparison of Jaccard similarity for patch token SAE features (threshold 0.001).

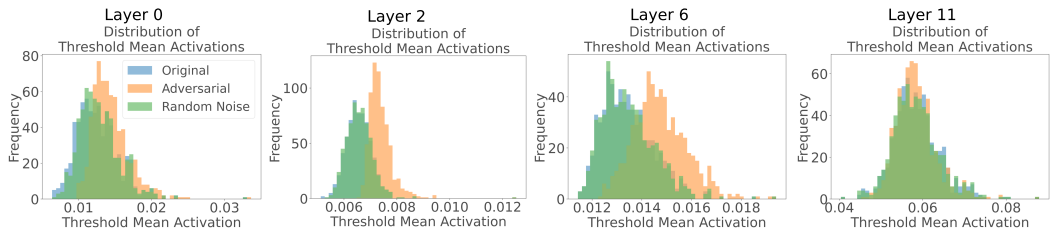


Figure 24: Histogram of mean activations for patch token SAE features (threshold 0.001), with noise.

CE cross entropy

LRH linear representation hypothesis

MLP Multilayer Perceptron

MSE mean squared error

NN neural network

PGD projected gradient descent

SAE sparse autoencoder

VIT vision transformer